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Reducing Overdose Deaths and Mitigating County Disparities: Optimal Allocation of Substance Use Treatment Centers

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Abstract. ***Problem definition*:** Substance use disorder (SUD) is a pressing health concern in the United States, and connecting communities with treatment is a growing public health imperative. We study the optimal allocation of substance use treatment centers across counties within U.S. states over a multiyear time horizon. We use a three-part objective function that minimizes (i) statewide overdose deaths, (ii) mismatch between counties' treatment center access and overdose death rates (inequity), and (iii) mismatch between counties' treatment center access and population (inequality). We also consider the temporal interdependence of counties' treatment center allocations and their death-rate-based equity targets. ***Methodology/results*:** We use a predict-then-optimize approach based on data from more than 2,500 U.S. counties between 1999 and 2016. We first estimate SUD treatment centers' causal impact on county-level overdose death rates, using mental health parity laws as an instrument. We then use these estimates in a nonlinear program that reallocates states' observed levels of SUD treatment funding. Results suggest that this approach can meaningfully reduce statewide overdose deaths and between-county disparities in treatment access. ***Managerial implications*:** Prioritizing death rate reductions over equity/equality results in more resources being directed to high-population counties, where treatment centers are most effective. Prioritizing equity/equality in treatment access results in more resources being directed to underserved, low-population counties. We also find that states differ in their Pareto-optimal trade-offs between death rates and inequity/inequality.

Supplemental Material: The online appendix is available at https://doi.org/10.1287/msom.2024.0963.

Keywords: substance use • resource allocation • healthcare operations • disparities

1. Introduction

Substance use disorder (SUD) is a growing public health challenge, and U.S. agencies have stepped up efforts to expand treatment access. SUD treatment is shown to reduce overdose rates (Stewart et al. 2002, Swensen 2015), reduce crime rates (Bondurant et al. 2018), and increase labor force attachment (McIntosh et al. 2008). Therefore, SUD treatment expenditures must be efficiently allocated among communities to maximize their impact (French and Drummond 2005). Yet there is considerable mismatch between SUD treatment access and SUD-related mortality in the United states (Langabeer et al. 2019). Such mismatches may be because of factors such as NIMBYism (i.e., neighborhood resistance to treatment center construction; Hunter 2022), difficulty staffing rural clinics (Lishner et al. 1996), or high costs in urban areas (McKay et al. 2008).

Because of the fast-changing nature of the SUD epidemic, policymakers and providers need empirically guided strategies for allocating SUD treatment resources. Building upon the literature in healthcare facility location and equitable resource allocation, we develop the first framework for optimizing SUD treatment centers among counties within U.S. states over a multiyear horizon. We use a predict-then-optimize approach to estimate the impact of SUD treatment centers on counties' drug-related overdose death rates and then allocate centers within U.S. states to minimize overdose deaths, serving as many communities as possible. Importantly, to guide policymakers in long-term planning, we model the temporal interdependence between communities' treatment resources, overdose deaths, and subsequent levels of treatment need. Our results suggest that optimization-driven treatment allocation

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can reduce statewide overdose deaths and improve equity and equality in treatment center allocation. Our findings also highlight the potential risk of increasing SUD treatment funding without data-driven guidance, namely, that allocating centers according to myopic heuristics may exacerbate both mortality and existing disparities in treatment access.

Practically, our results are relevant to policymakers and SUD treatment providers. When there are disparities in SUD treatment access, state governments can strategically guide resource allocation within their borders (Purrington 2019) given their role in administering federal SUD treatment block grants (Substance Abuse and Mental Health Services Administration 2024). Such grants totaled $1.5 billion in 2021 and represent roughly 40% of public SUD treatment spending. Our results are also important to healthcare providers. Many states have certificate of need laws (Conover and Bailey 2020) under which providers must demonstrate community need before opening new facilities. Therefore, understanding counties' optimal numbers of treatment centers can help providers make and defend decisions on service expansion. Thus, by formalizing SUD treatment facility placement as an optimization problem, we develop a data-driven framework for quantifying counties' SUD treatment need, which can inform both policy and healthcare decision making.

2. Literature Review

SUD is "a mental health condition in which a person has a problematic pattern of substance use that causes distress and/or impairs their life" (Cleveland Clinic 2024), and can involve substances such as alcohol, opioids, stimulants, and hallucinogens. Treatment can include residential treatment, outpatient treatment, medication-assisted treatment, or some combination. Research shows that treatment reduces substance use and relapse (Stewart et al. 2002) and improves other measures of social health, including overdoses (Swensen 2015), local crime (Bondurant et al. 2018), and employment (McIntosh et al. 2008).

2.1. Optimization-Driven Healthcare Facility Allocation

The study of optimal healthcare facility location has a rich history in operations management (Rais and Viana 2011, Ahmadi-Javid et al. 2017). Ahmadi-Javid et al. (2017) provide a comprehensive review of the healthcare facility location problem, including applications to primary care, organ transplant centers, and mobile health clinics. In such studies, a fixed number of facilities can be placed to maximize coverage, minimize travel distance, minimize the total cost of facilities, or assign patients to existing facilities to minimize wait times.

The use of coverage or travel time to evaluate allocations is highly relevant for emergency services and for contexts in which resources are extremely limited, such as rural areas in developing countries. In contrast, we argue that SUD treatment allocation in the United States requires a different framework. In 2016, the United States had more than 18,000 SUD treatment centers (or about six centers for each of the roughly 3,000 U.S. counties), and 95% of the U.S. population had at least one treatment center in the county. Therefore, the primary research question for SUD allocation is not adequate coverage (i.e., ensuring all residents have access to at least one center), but how the number of centers in each county aligns with community needs. We argue that centers should be allocated to optimize objective health and recovery outcomes and, specifically, drug-related overdose deaths. Because research demonstrates a significant relationship between treatment access and drug-related overdose deaths (Stewart et al. 2002, Swensen 2015), this allows us to quantify the efficiency of treatment center allocations in terms of a single public health metric in contrast with past studies that focus on coverage or cost (Ahmadi-Javid et al. 2017).

Studies that do allocate healthcare resources to minimize mortality typically focus on vaccine distribution (Medlock and Galvani 2009, Buhat et al. 2021). To our knowledge, Long et al. (2018) is the only healthcare facility location study that directly optimizes an objective public health outcome (disease infection rates) rather than geographic coverage or travel time. Long et al. (2018) uses a susceptible–infected–recovered infectious disease model to simulate Ebola infection rates under various allocations of treatment units. By estimating the number of averted infections for each treatment bed, the authors quantify the degree to which various allocation strategies reduce infections. We use a similar approach for allocating SUD treatment centers to minimize drug-related mortality. However, the lack of well-established models for drug-related overdose deaths (in contrast with well-established infectious disease models) adds complexity to our analysis and demonstrates the value of this approach for conditions besides infectious diseases. Somewhat similarly, Jakubik and Feuerriegel (2022) use a predict-then-optimize approach to allocate international aid to minimize HIV infections though this study focuses on aid disbursement rather than facility allocation.

2.2. Incorporating Equity and Equality into Allocation Decisions

Alongside the efficiency of a given SUD resource allocation strategy (i.e., its impact on mortality), decision makers must also consider whether resources are equitably distributed. There is a robust literature on this efficiency–fairness trade-off in which fairness is defined as an even distribution of utility across decision units (Bertsimas et al. 2012, Karsu and Morton 2015, Argyris et al.

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2022). In our context, this encourages allocations that equalize overdose mortality across counties. Yet this approach suffers drawbacks in the context of SUD. First, counties vary widely in their overdose mortality rates, and using treatment centers to correct existing discrepancies may yield highly unequal treatment center distributions. Second, equalizing mortality rates across counties of different populations (at the expense of overall mortality reduction) implies that some lives are worth more than others—an undesirable assumption for policymakers.

We, thus, consider whether a given allocation distributes treatment access in an equitable manner. We specifically build upon Rea et al. (2021), who argue that decision makers should consider both equality (when resources are evenly distributed) and equity (when resources are distributed according to merit, need, or importance). The authors formalize inequality as the L2-norm of the deviations between the proposed allocation and a perfectly equal allocation and inequity as the L1-norm of these deviations weighted by some measure of importance. Equity and equality are therefore crucial considerations in our optimization given that SUD disproportionately affects under-resourced communities. Of course, allocating centers purely based on equality/equity may be unsustainable and impact centers' financial viability. We therefore balance these objectives in a framework that allows decision makers considerable flexibility on how to prioritize death rates, equity, and equality. Balancing allocation efficiency with equality/equity also sets our work apart from Long et al. (2018) and Jakubik and Feuerriegel (2022), who focus exclusively on efficiency.

2.3. Sequential Consideration of Equitable and Equal Resource Allocation

Because policymakers should consider SUD mitigation efforts over multiyear horizons, our framework must account for sequential allocation equity and equality. A few studies address fair resource allocation in sequential contexts. Ma et al. (2023) and Lien et al. (2014) consider multiperiod nonprofit resource allocation and quantify fairness as the maximum demand shortfall across units. Similarly, Firouz et al. (2023) optimize food bank distributions across agencies, incorporating fairness by penalizing agencies' deviations from the maximum demand fill rate. Sun et al. (2023) optimize shift scheduling for anesthesiologists by evenly distributing difficult/undesirable shifts across physicians. Similarly, Gross et al. (2019) optimize physician shift scheduling over a multimonth horizon and develop a metric for ensuring that certain physicians' preferences are not repeatedly violated. Notably, in these studies, resource targets or demands are known beforehand or vary according to a known distribution and are, therefore, exogenous to the decision model. As we discuss in Section 3.4.2, we argue that equitable SUD treatment allocation should consider counties' SUD overdose death rates, which endogenously depend on prior treatment allocations. Thus, in our context, decision units' deviations from their resource targets and the resource targets themselves depend on prior allocation decisions. This constitutes a core contribution of our framework. The equity/equality resource allocation approach pioneered by Rea et al. (2021) has not (to our knowledge) been extended to the sequential setting in which resource targets depend on prior allocations.

2.4. Prior Data-Driven Approaches to SUD Treatment Decision Making

A few prior studies leverage data-driven techniques to study SUD treatment. Baucum et al. (2023) develop a machine learning (ML) algorithm for matching SUD patients with treatment programs, whereas Nasir et al. (2021) and Lee et al. (2019) use ML to predict patients' likelihood of SUD treatment dropout and treatment-seeking behavior, respectively. Tragler et al. (2001) develop a system dynamics model of drug epidemics that predicts the total number of users as a function of treatment and enforcement expenditure. Still, there has been no research studying how limited SUD treatment resources should be distributed across communities to improve outcomes. The most relevant study is Langabeer et al. (2019), which quantifies SUD treatment availability relative to counties' overdose rates and finds that almost one in five U.S. counties are underserved with respect to this metric. Yet identifying strategies for allocating treatment centers in a way that improves access and reduces overdose mortality is an unstudied research question.

2.5. Summary and Proposed Contributions

Our study contributes both theoretically and practically to the literature on this critical but understudied topic. To our knowledge, this is the first study to investigate how SUD treatment resources can be allocated to minimize drug-related mortality, ensuring equitable access over a multiyear horizon. We build on the work of Long et al. (2018) and Jakubik and Feuerriegel (2022) (which allocate resources to minimize infection rates) and the elegant Rea et al. (2021) framework for equal and equitable resource allocation. In building on these studies, our work is unique in its consideration of equity and equality alongside empirical estimates of allocation effectiveness. Furthermore, we consider temporal dynamics in a multiyear context (in contrast with Rea et al. 2021, Jakubik and Feuerriegel 2022) by modeling the temporal interdependence between an objective public health metric and local treatment need and how such interdependence impacts resource allocations. Practically, our work offers a tool that policymakers and public health entities may use to combat the U.S. SUD epidemic in a more efficient, equitable, and equal manner.

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3. Methods

3.1. Data Set

We obtain county-specific counts of SUD treatment centers from the U.S. Census Bureau's County Business Patterns database for 1999–2016. Treatment centers are denoted by North American Industry Classification System codes 621420 for outpatient treatment centers and 623220 for residential treatment centers; we calculate each county's treatment center count by summing across residential and outpatient centers. Following past research (Swensen 2015, Bondurant et al. 2018), we limit our analysis to the contiguous 48 U.S. states. Treatment center counts are only collected through 2016 because counties with fewer than three treatment centers are excluded starting in 2017.

Our outcome is log-transformed drug-related mortality rates in U.S. counties, which we obtain from the Centers for Disease Control (CDC) Multiple Causes of Death database for 1999–2017 (one year later than our treatment center data because our econometric model uses a one-year lag). We define drug-related mortality as county deaths per 100,000 residents because of drug overdose or alcohol poisoning (International Classification of Diseases, 10th revision codes X40-X45, X60-X65, X85, and Y10-Y15). We also control for counties' changing demographic and economic conditions over time. All variables appear in Table 1. Note that employment–population ratios were missing for 11 county–year records and were forward-filled from prior years. Opioid prescription rates (Drug Enforcement Agency 2024) are originally available for three-digit ZIP codes, which we map to counties for analysis (whereas we focus on SUD across all substances, we control for opioid prescriptions because of the historical role they have played in the U.S. SUD epidemic). We also log-transform county population and opioid prescription rates because of high positive skewness. Variable citations appear in Online Appendix EC.1.

3.2. Instrumental Variables for County-Level Treatment Center Counts

We use an instrumental variable (IV) approach to estimate the causal impact of SUD treatment centers on overdose death rates, using states' mental health parity laws as our primary instrument.

Mental health parity is the legal requirement that insurance benefits for mental health conditions (including SUD) are comparable to those offered for other medical conditions (Wen et al. 2013). In the absence of explicit parity requirements, "coverage for SUD treatment has been more restrictive than that for medical/surgical treatment" (Wen et al. 2013, p. 1356). The increased reimbursements required by parity legislation should increase demand for and profitability of SUD treatment, therefore meeting the relevance condition for valid instruments. Previous studies have, therefore, exploited variation in parity laws to examine SUD treatment access across states and over time (Dave and Mukerjee 2011, Wen et al. 2013).

The validity of parity laws as an IV also rests on the assumption that they only affect overdose deaths through their impact on SUD treatment center counts (the exclusion condition) conditional on controls. By themselves, mental health parity laws do not affect the pathways toward overdoses other than by increasing the financial viability of treatment as they do not entail primary prevention efforts. We, therefore, exploit the exogenous variation in health insurance coverage induced by parity laws to examine the impact of treatment centers on overdose death rates. This is comparable to studies using Medicaid expansion under the Affordable Care Act to examine the effect of health insurance coverage on outcomes such as mortality (Sommers et al. 2012) and crime (He and Barkowski 2020). Unlike Medicaid expansion, mental health parity laws are not accompanied by campaigns to drive enrollment and engagement with therapy. Therefore, mental health parity laws are unlikely to affect overdose deaths through channels outside of improved access to SUD treatment and the financial viability of treatment centers. Furthermore, by controlling for community factors, such as employment, per-capita income, and poverty rate, we ensure that any effects of parity are not attributable to broader economic spillovers from such policies. Because we control for states' uninsured rates, the effects of parity laws should not be attributable to any correlation with overall insurance coverage rates. We also control for states' share of SUD patients using private insurance to ensure the effect of parity laws is not attributable to improved payment options among existing SUD patients.

States' parity laws can be classified as requiring no parity, partial parity (mental health coverage is mandated but may be less generous than medical coverage), parity-if-offered (mental health coverage must be equal to medical coverage if it is offered), or full parity (mental health coverage is mandated and must be equal to medical coverage) (Wen et al. 2013, National Conference of State Legislatures 2015). We specifically focus on states' mental health parity requirements for large-group private health insurance (usually defined as employers with ≥ 50 employees) because (i) there is meaningful between-state variation over our study's time period (1999–2016) and (ii) small-group and individual plans are all subject to full parity starting in 2010 because of the Patient Protection and Affordable Care Act (Frank et al. 2014). The 2008 Mental Health Parity and Addiction Equity Act (MHPAEA) mandated parity-if-offered for all large-group plans; thus, before the MHPAEA's passage (1999–2008 in our data set), large-group insurers could be subject to no parity, partial parity, parity-if-offered, or full parity requirements, depending on state law. After the law's passage (2009–2016 in our data

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Table 1. Variables Used for Analysis

Variable	Units	Level	Notation
County index (from one to M counties for a given state)	/	County	$m \in \{1, \ldots, M\}$
Year index (from year t_0 over H-year horizon)	/	Year	$t \in \{t_0, \ldots, t_0 + H - 1\}$
Drug-related overdose deaths	Deaths/100,000 residents	County	$y_{t.m}$
SUD treatment centers	Number of centers	County	$x_{t.m}$
Parity laws	Dummies	State	$z_{t.m}^{ofr}, z_{t.m}^{prt}, z_{t.m}^{fll}$
Population/100,000	Number of residents in 100,000s	County	$c_{t.m}^{(1)}$
Log(population)	Log(number of residents)	County	$c_{t.m}^{(2)}$
Per-capita income	U.S. dollars, in thousands	County	$c_{t.m}^{(3)}$
Employment–population ratio	Percentage	County	$c_{t.m}^{(4)}$
Poverty rate	Percentage	County	$c_{t.m}^{(5)}$
Rurality	Percentage	County	$c_{t.m}^{(6)}$
Percentage of population 19 years or younger	Percentage	County	$c_{t.m}^{(7)}$
Percentage of population 65 years or older	Percentage	County	$c_{t.m}^{(8)}$
Log(opioid prescription rates)	Log(morphine kg equivalent/100,000 residents)	County	$c_{t.m}^{(9)}$
Medicaid coverage expanded	Binary (0/1)	State	$c_{t.m}^{(10)}$
Uninsured rate	Percentage	State	$c_{t.m}^{(11)}$
Percentage SUD patients paying with private insurance	Percentage	State	$c_{t.m}^{(12)}$

Note. SUD treatment centers' emphasized as key explanatory variable.

set), large-group insurers could be subject to parity-if-offered (in the absence of other mental health parity laws) or full parity (if states mandate coverage of mental health benefits).

We catalog parity laws and implementation dates for all 48 contiguous U.S. states from 1999 to 2016. Implementation dates are triangulated based on previous analyses by Wen et al. (2013), the National Conference of State Legislatures (2015), and ParityTrack (2021) and additional analysis of state legislative texts when necessary. Our catalog of parity laws appears in Online Appendix EC.2. To allow the effect of parity legislation (in terms of number of centers opening/closing) to vary by county size, we interact our parity dummy variables with lagged county population (controlling for county population) in line with other studies that interact instruments with measures of county/area size to allow for heterogeneous policy effects (Eriksen and Rosenthal 2010, Rosenblum et al. 2017). Finally, because parity can be enacted at the state or federal level, we confirm that states that enacted their own parity laws do not meaningfully differ from others in their preparity number of treatment centers. We find that, in the five years prior to each state's first parity law change (whether state or federal), states with their own parity laws had an average of 5.92 centers per 100,000 residents (SD = 1.82) compared with 5.56 (SD = 1.82) among states where only federal parity laws took effect (visualized in Online Appendix EC.3).

Finally, we include two supplemental analyses to further validate our instrument set and empirical estimates. First, Online Appendix EC.4 presents a placebo test showing that mental health parity laws had no significant impact on cancer-related mortality. This suggests that any empirical effect of mental health parity laws on overdose mortality is not because of their general impact on healthcare-seeking behavior (in which case, non–SUD-related medical outcomes would also be affected). Second, Online Appendix EC.5 reestimates our empirical model with an alternative IV set, which yields a very similar coefficient estimate (within 0.28 standard error) and further validates the estimate from our parity-based analysis.

3.3. Estimating Treatment Center Effects on Mortality

We now model the effect of SUD treatment centers on log-transformed overdose deaths; the log-transformation reflects the finding that additional centers have diminishing effects on reducing county-level overdose mortality (Tragler et al. 2001). This follows past research on overdose death rates (Pardo 2017, Marotta et al. 2019) and reduces skewness in the outcome (untransformed skewness of 2.31, transformed skewness of −0.37). We use county- and year-level fixed effects to account for unobserved time-invariant differences between counties and location-invariant characteristics of each year. Counties are population weighted (Swensen 2015), and standard errors are clustered at the state level.

We define the log-transformed overdose death rate (per 100,000 residents) in county m in year t as $\log(y_{t.m})$, and model the effect of treatment centers on $\log(y_{t.m})$ using two-stage least squares:

$$\hat{x}_{t.m} = \dot{\beta}_1 z_{t.m}^{ofr} + \dot{\beta}_2 z_{t.m}^{prt} + \dot{\beta}_3 z_{t.m}^{fll} + \dot{\beta}_4 z_{t.m}^{ofr} c_{t.m}^{(1)} + \dot{\beta}_5 z_{t.m}^{prt} c_{t.m}^{(1)} + \dot{\beta}_6 z_{t.m}^{fll} c_{t.m}^{(1)} + \sum_{k=1}^{K} \dot{\beta}_k c_{(t+1).m}^{(k)} + \nu_t + \nu_m, \quad (1)$$

$$\log(\hat{y}_{(t+1).m}) = \theta \hat{x}_{t.m} + \sum_{k=1}^{K} \beta_k c_{(t+1).m}^{(k)} + \mu_{t+1} + \mu_m, \quad (2)$$

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where $x_{t.m}$ (our main explanatory variable of interest) is the number of centers in county m in year t; $z^{ofr}_{t.m}$, $z^{prt}_{t.m}$, and $z^{fll}_{t.m}$ are dummies for parity-if-offered, partial parity, and full parity (respectively); $c^{(k)}_{t.m}$ for $k \in \{1, \ldots, K\}$ are values for the $K = 12$ control variables from Table 1 (with $c^{(1)}_{t.m}$ representing county population); and $(\nu_t, \nu_m, \mu_t, \mu_m)$ are year and county fixed effects for stage 1 (ν) and stage 2 (μ). Note that parity dummies are equal to one for the first full year after law passage, following Wen et al. (2013); thus, we assume that parity legislation can only affect SUD treatment centers in the year after they are passed.

We follow past research (Swensen 2015) in using treatment center counts to predict changes in death rates. This implies that a single treatment center averts more overdose deaths in high-population counties than in low-population counties, which is supported by past studies (Pullen and Oser 2014, Edmond et al. 2015, Ghose et al. 2022). This is also supported by the fact that treatment utilization is higher in urban (versus rural) areas (Warner and Leukefeld 2001) and that urban centers have greater capacity than rural centers (which we account for in Section 3.5.2).

3.4. Optimizing SUD Treatment Center Locations

For a given U.S. state with m counties, we assume that government agencies and/or healthcare decision makers may wish to optimally allocate treatment centers over an H-year horizon $\{t_0, \ldots, t_0 + H - 1\}$ to minimize statewide overdose deaths in following years $\{t_0 + 1, \ldots, t_0 + H\}$ and to ensure fair allocation of SUD treatment resources among counties. Section 3.4.1 formalizes the objective to minimize statewide overdose deaths, Sections 3.4.2 and 3.4.3 formalize the objective to ensure fair SUD treatment allocation among counties, and Section 3.4.4 formalizes the joint optimization problem.

3.4.1. Objective for Minimizing Overdose Deaths. For a given center allocation in year t, county m's overdose death rate in year $t + 1$ is estimated by $\hat{y}_{(t+1).m} = \exp(f(x_{t.m}, \mathbf{c_{(t+1).m}}, \mu_m, \mu_{t+1}) + \sigma^2/2)$, where $f()$ is the econometric model from Equation (2) and the $\exp(\ldots + \sigma^2/2)$ (known as the Smearing retransformation; Duan 1983) converts the log-transformed prediction to the predicted mean in original units, in which σ is the residual standard deviation. Thus, for a given U.S. state, if treatment center counts $\mathbf{x}_t$ are allocated among M counties with populations $\mathbf{c}^{(1)}_t$ starting in year t, then the model-estimated statewide average overdose death rate in year $t + 1$ is given by

$$J_t(\mathbf{x}_t) = \frac{1}{M}\sum_m \exp(f(x_{t.m}, \mathbf{c_{(t+1).m}}, \mu_m, \mu_{t+1}) + \sigma^2/2) \times (c^{(1)}_{(t+1).m}). \quad (3)$$

3.4.2. Ensuring Equity and Equality in SUD Treatment Center Allocation. Along with minimizing the statewide overdose death rates, policymakers may also wish to ensure some degree of equity and/or equality in counties' access to SUD treatment; this is in line with recent calls by academics and practitioners to balance equity and efficiency in healthcare resource allocation (Asamani et al. 2021). We, thus, draw on recent research on fair resource allocation (Rea et al. 2021) to introduce SUD treatment equality and equity into our optimization. According to Rea et al. (2021), equality is satisfied when resources are evenly distributed across units (such as employees receiving equally sized bonuses), whereas equity is satisfied when resources are allocated in proportion to a unit's inputs (such as employees receiving bonuses proportional to their performance ratings). Given our focus on treatment, we instead follow the healthcare literature (Diderichsen 2004) and define equity as allocating resources in proportion to communities' need (in contrast with equality, in which healthcare resources are evenly distributed regardless of differential need). Thus, for our purposes, treatment centers are equally distributed when counties' shares of statewide treatment centers equals their shares of statewide population (i.e., when $x_{t.m} = (\sum_j x_{t.j})(c^{(1)}_{t.m}/\sum_j c^{(1)}_{t.j})$). Treatment centers are equitably distributed when counties' shares of statewide treatment centers equals their shares of statewide overdose deaths (i.e., when $x_{t.m} = (\sum_j x_{t.j})(y_{t.m}c^{(1)}_{t.m}/\sum_j y_{t.j}c^{(1)}_{t.j})$).

We quantify inequity and inequality in treatment center allocation with functions Q^{eqt} and Q^{eql}:

$$Q^{eqt}_t(\mathbf{x}_t) = \frac{1}{M}\left\| \frac{1}{c^{(1)}_{t.m}}\left(x_{t.m} - \sum_j x_{t.j}\frac{y_{t.m}(c^{(1)}_{t.m})}{\sum_j y_{t.j}(c^{(1)}_{t.j})}\right)\right\|_{p_{eqt}}, \quad (4)$$

$$Q^{eql}_t(\mathbf{x}_t) = \frac{1}{M}\left\| \frac{1}{c^{(1)}_{t.m}}\left(x_{t.m} - \sum_j x_{t.j}\frac{c^{(1)}_{t.m}}{\sum_j c^{(1)}_{t.j}}\right)\right\|_{p_{eql}}, \quad (5)$$

where M is the number of counties in the state, $c^{(1)}_{t.m}$ is population (divided by 100,000) for county m in year t, $x_{t.m}$ is the number of treatment centers for county m in year t, and $\sum_j x_{t.j}$ is the total number of centers statewide. The p_{eqt} and p_{eql} norms in each equation are computed over the M counties in the state. Thus, the difference term in parentheses is the discrepancy between (i) the number of centers in each county and (ii) that county's proportion of statewide deaths (inequity) or population (inequality) times the number of statewide centers. This difference is then normalized by county population $c^{(1)}_{t.m}$. In other words, inequity and inequality are the p_{eqt} and p_{eql} norms of county-level deviations from perfectly equitable/equal treatment center distribution in terms of treatment centers per 100,000 residents. Finally, note that Rea et al. (2021) uses a single target allocation across the inequity and inequality functions with the inequity function differentially weighting discrepancies from the target. In contrast, we use separate targets for each function because

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our healthcare focus affords an interpretable target allocation under both inequality (population-based allocation) and inequity (death rate–based allocation).

3.4.3. Equity Target Dependence on Prior Allocation Decisions. When optimizing treatment center allocations over an H-year time horizon $\{t_0, \ldots, t_0 + H - 1\}$, the inequity function Q_t^{eqt} above is only valid for the first year, $t = t_0$. For later years, $t > t_0$, using observed county-specific death rates $y_{t.m}$ ignores the impact of prior allocations on SUD death rates. Thus, counties' equity targets in year $t > t_0$ depend on decisions made in years $t_0, \ldots, t-1$. We, therefore, use our econometric model to predict counterfactual death rates under a given center allocation for years $t > t_0$. Thus, when $t > t_0$, we denote the inequity function as $\tilde{Q}_t^{eqt}(\mathbf{x}_t)$, which has the following form:

$$\tilde{Q}_t^{eqt}(\mathbf{x}_t) = \frac{1}{M} \left\| \frac{1}{c_{t.m}^{(1)}} \left(x_{t.m} - \sum_j x_{t.j} \frac{\exp\left(f(x_{(t-1).m}, \mathbf{c_{t.m}}, \mu_m, \mu_t) + \frac{\sigma^2}{2} \right) c_{t.m}^{(1)}}{M \cdot J_{(t-1)}(\mathbf{x}_{(t-1)})} \right) \right\|_{p_{eqt}}, \tag{6}$$

where $M \cdot J_{(t-1)}(\mathbf{x}_{(t-1)})$ is the estimated number of statewide deaths in year $t-1$ computed by summing the numerator across all M counties. This function introduces nonlinearity into counties' inequity scores; adding centers to an underserved county both increases its center count and decreases its equity target.

3.4.4. Multiyear Optimization Formula. Our optimization combines $J()$, which quantifies efficiency in terms of mortality outcomes, with $Q^{eqt}()$ and $Q^{eql}()$, which quantify inequity and inequality in terms of treatment access. Thus, for a given U.S. state in year t_0 with M counties, we seek to allocate treatment centers over the next H years $\{t_0, \ldots, t_0 + H - 1\}$ according to

$$\begin{aligned} \min_{\mathbf{x}_{t_0}, \cdots, \mathbf{x}_{t_0+H-1}} \frac{1}{H} \; & (\alpha \cdot J_{t_0}(\mathbf{x}_{t_0}) + (1-\alpha)[\lambda \cdot Q_{t_0}^{eqt}(\mathbf{x}_{t_0}) \\ & + (1-\lambda) \cdot Q_{t_0}^{eql}(\mathbf{x}_{t_0})] + \sum_{t=t_0+1}^{t_0+H-1} \alpha \cdot J_t(\mathbf{x}_t) \\ & + (1-\alpha)[\lambda \cdot \tilde{Q}_t^{eqt}(\mathbf{x}_t) + (1-\lambda) \cdot Q_t^{eql}(\mathbf{x}_t)]) \\ \text{s.t.} \quad & \sum_m v_m(x_{t.m}) + v^{(fixed)}(d_{t.m}) \le B_t \quad \forall t \\ & x_{t.m} \ge L * x_{t_0.m}^{(actual)} \quad \forall m, t \\ & x_{t.m} \text{ integer} \quad \forall m, t \\ & d_{t.m} \ge x_{t.m} - x_{(t-1).m} \quad \forall m, t \\ & d_{t.m} \ge 0 \quad \forall m, t, \end{aligned} \tag{7}$$

where

- $\alpha \in [0,1]$ is the weight placed on overdose death minimization (versus inequity/inequality), $\lambda \in [0,1]$ is the equity (versus equality) weight, v_m is the cost of operating a center in county m, and B_t is the allowed budget for year t.
- $v^{(fixed)}$ is the cost of opening a new center, and $d_{t.m}$ is the number of centers opened in year t in county m. Because $d_{t.m}$ detracts from the available budget, it takes on its smallest allowed value ($x_{t.m} - x_{(t-1).m}$ if $x_{t.m} \ge x_{(t-1).m}$ or zero otherwise) as long as additional centers are beneficial; the nonnegativity constraint on $d_{t.m}$ ensures that opening costs are not recouped when centers are closed.
- $L \in [0,1]$ is a ratio that determines the minimum proportion of centers that must remain in each county relative to its observed count in baseline year t_0. That is, the proportion of centers that can be removed from each county cannot exceed $(1-L)$. Setting $L = 1$ requires that centers only be added to each county, whereas $L = 0$ places no restriction on how many centers can be closed/removed in each county. We advocate for setting $0 < L < 1$, which allows resources to be reallocated away from existing centers without decreasing counties' center counts to an unreasonable degree based on decision makers' preferences. Note that setting $L > 0$ also prevents solutions that only minimize inequity and inequality from assigning zero centers to all counties. We further justify this parameter and provide a supplemental analysis for the case of $L = 0$ in Online Appendix EC.6.

Because the only lower bound on each $x_{t.m}$ is $L \cdot x_{t_0.m}^{(actual)}$, the optimization can decrease a given county's treatment center count (i.e., recommend center closure) up to a proportional reduction of $(1-L)$. This allows operational funds from those centers to be redirected to other counties but requires spending $v^{(fixed)}$ to construct additional centers elsewhere (which cannot be recouped from the closed center).

Following Rea et al. (2021), we use $p_{eqt} = 1$ and $p_{eql} = 2$ for computing inequity and inequality, and we square the L2 inequality norm to solve the problem using quadratic programming. The L2 norm for Q^{eql} is necessary to discourage extreme inequality in counties' treatment center densities. In contrast, the L1 inequity norm is beneficial because counties' overdose deaths fluctuate to a greater degree than population (average within-county equity target SD = 2.07 compared with 1.07 centers for equality). Thus, Q^{eqt} does not penalize large equity deviations (which may correct themselves through fluctuations in death rates) as strongly as Q^{eql} penalizes large equality deviations, which are more likely to persist into future years. In Online Appendix EC.7, we show that the combined equity–equality target for each county lies in between its equity and equality targets and visualize how counties' combined center targets depend on λ.

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3.5. Optimization Analysis Parameters and Procedure

We use data from 1999–2016 to estimate the instrumental variable regression model in Equations (1) and (2). We then run our optimization for the 48 continental U.S. states using base year $t_0 = 2016$ over an $H = 4$-year time horizon, such that treatment center allocations are optimized for 2016, 2017, 2018, and 2019. We choose $H = 4$ given the four-year length of state and federal administrations and given similar timelines among federal and state SUD mitigation strategies (Mishra and Mathias 2019, Office of National Drug Control Policy 2022). Note that 2019 solutions are evaluated with respect to overdose deaths in 2020, which were elevated nationwide because of the COVID-19 pandemic. The inclusion of 2019 as a decision year is intentional to assess our method's performance amid unexpected rises in SUD mortality. Though, because we present results on a year-by-year basis (e.g., Figure 2), we show that our results are not driven by the use of 2020 data. Online Appendix EC.8 reruns our results using pessimistic worst case forecasts of nationwide SUD death rates, which do not offer meaningful improvement over our primary results.

3.5.1. Use of Estimated vs. Actual County Covariates and Fixed Effects. The objective components $J_t()$ (for all t) and $\tilde{Q}_t^{eqt}, Q_t^{eql}$ (for $t > t_0$) rely on covariates $\mathbf{c}_{t.m}$ and year fixed effects μ_t that are not known in decision year t_0. When solving the optimization, we use forecasted values of county covariate vectors $(\mathbf{c}_{2017.m}, \ldots, \mathbf{c}_{2020.m})$ and year-specific fixed effects $(\mu_{2017}, \ldots, \mu_{2020})$ based on 1999–2016 data (Lien et al. 2014, Long et al. 2018) as follows:

- For year fixed effects $(\mu_{2017}, \ldots, \mu_{2020})$, we forecast forward from μ_{2016} using the average linear trend from $(\mu_{1999}, \ldots, \mu_{2016})$. Whereas this method accurately captures the increase in nationwide overdose deaths from 2017 to 2020, it underestimates μ_{2020} (which was elevated because of COVID-19); this allows us to assess whether our approach performs well even when overdose deaths increase unexpectedly.
- For county population, we forecast forward to 2017–2020 based on the county's average growth rate from 1999 to 2016. For county rurality (which is only available from the decennial census) and Medicaid expansion, we forward-fill values for 2017–2020.
- For all other covariates, we train random forests (maximum depth of five with 100 estimators) using 1999–2016 data to predict covariates in year t based on years $(t-10, \ldots, t-1)$, and use the trained models to predict counties' 2017–2020 covariates in a rolling fashion. The average correlation between predicted and actual values is $\bar{r} = 0.91$.

After solving our optimization using the estimated values above, each solution is evaluated using ground-truth, held-out county-level covariates $\mathbf{c}_{2017}, \ldots, \mathbf{c}_{2020}$ and year fixed effects $\mu_{2017}, \ldots, \mu_{2020}$, according to $J_t()$, $Q_t^{(eqt)}$, and $Q_t^{(eql)}$ for $t \in \{2017, \ldots, 2020\}$.

3.5.2. Treatment Center Cost v_m, Fixed Cost $v^{(fixed)}$, and Budgets B_t. The cost of each treatment center is $v_{t.m} = \overline{n}_{t.m}\overline{h}_{t.m}$, where $\overline{n}_{t.m}$ is the predicted average number of employees per treatment center in county m and $\overline{h}_{t.m}$ is the average yearly salary for education and health workers in county m in year t (taken from Bureau of Labor Statistics 2024). Note that education and health is the Bureau of Labor Statistics industry designation most closely related to SUD treatment.

We find that centers in high-population areas generally have more employees than those in low-population counties, according to $\overline{n}_{t.m} = 21.55 + 5.61 \log(c_{t.m}^{(1)})$. If employee counts are a proxy for center capacity, this partly explains why centers in high-population counties would avert more overdose deaths than those in low-population counties. We also find that education and health services salaries $(\overline{h}_{t.m})$ in high-population counties are higher than those in low-population counties with median salaries in the top half of counties (by population), 39% higher than those in the bottom half. Thus, $\overline{h}_{t.m}$ and $\overline{n}_{t.m}$ both imply that treatment centers in high-population counties require more resources than those in low-population counties. We base $v_{t.m}$ on treatment centers' wage expenditures because labor constitutes the majority of SUD treatment centers' operating expenses (French et al. 1997); start-up capital costs are accounted for with $v^{(fixed)}$.

For $v^{(fixed)}$, we rely on start-up estimates of \$200,000–\$400,000 (Drug Rehab Agency 2022) and \$300,000–\$600,000 (Ascension Recovery Services 2022), both from SUD treatment consultancies, for an average estimate of $v^{(fixed)} = \$375,000$. We test our results' sensitivity to this value in Online Appendix EC.9. For context, this estimate is higher than the estimated \$100,000 startup costs for primary care offices (Palmer 2020, Freedman 2024), likely because of some centers' use of residential facilities and the wide array of social services that some centers offer to clients.

Finally, we define the treatment center budget B_t for 2016–2019 based on states' actual increases in treatment expenditures. For each year t, we set $B_t = \sum_m v_m(x_{t.m}^{(actual)}) + \sum_m \max[v^{(fixed)}(x_{t.m}^{(actual)} - x_{(t-1).m}^{(actual)}), 0]$, that is, the cost of operating each states' actual center allocations plus the fixed cost of any centers opened that year. Because center counts below three are suppressed for years 2017–2020, we forward-fill missing center counts from 2016 (note that applying this same procedure to 2013–2016 yields an average discrepancy of just 0.7% in nationwide center counts). Thus, our optimization specifically tests whether resources that states have already committed to SUD treatment could have been allocated

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in a way that further reduced death rates or inequity/inequality.

3.5.3. Weights λ and α. We use $\lambda = 0.5$ for our primary analysis; that is, we place equal weight on the inequality and inequity objectives. As we show in Online Appendix EC.7, placing equal weight on the inequity and inequality metrics yields solutions that are closer to counties' equality targets than their equity targets because inequality uses a nonlinear L2 norm. This reflects our intent to penalize equality deviations more aggressively than equity deviations because the latter are more likely to be the result of high variability in county-level overdose deaths. We present results for several other values of λ in Section 4.3.2 and Online Appendix EC.10 to further highlight the impact of the λ parameter. Finally, the weight α relates the utility of preventing one overdose death per county to reducing the inequity/inequality metric by one; we test several values of $\alpha \in \{0, 0.1, 0.25, 0.5, 0.75, 0.9, 1\}$. Note that the case of $\alpha = 0$ is equivalent to the framework from Rea et al. (2021). This case can, therefore, serve as an additional benchmark for our proposed framework in which efficiency is incorporated into the objective function ($\alpha > 0$).

3.5.4. Benchmarking. We compare our optimal solutions with states' actual treatment center allocations (i.e., those used to compute the budgets B_t). We also benchmark against three myopic allocation strategies that policymakers might feasibly adopt that do not rely on our optimization framework:

- We benchmark against a proportional increase solution in which treatment center counts in each county are increased by the same proportion in a given year (this is solved via a linear program such that the total expenditures are as close to the budget B_t as possible). We allow for fractional treatment centers because rounding often produces nonincreases for small counties.
- We benchmark against a population-based solution in which states' budgets B_t are allocated according to county population from the prior year (e.g., a county with 10% of the statewide population receives 10% of B_t). For each county, its yearly budget is first spent to operate existing centers and then (if funds are available) to open new centers. If a county has unused funds that are not sufficient to open and operate a new center, they are rolled over to the following year.
- We benchmark against a death-based solution in which B_t is allocated according to counties' overdose deaths from the prior year (e.g., a county with 10% of statewide overdose deaths receives 10% of B_t) with funds spent according to the same procedure as the population-based benchmark.

Finally, we also compare cases in which $\alpha > 0$ with the special case of $\alpha = 0$, which only considers inequity and inequality and is equivalent to the framework proposed by Rea et al. (2021). All solutions and benchmarks are evaluated using the death rate, inequity, and inequality components of the objective function in Equation (7), using held-out, ground-truth data from years 2017–2020.

3.5.5. Optimization Solution Algorithm. We utilize two relaxations to solve our optimization. First, the optimization is nonconvex because of the presence of $J()$ (which depends on the decision variables $\mathbf{x}_{t.m}$) in the denominator of $\tilde{Q}_t^{eqt}$. To address this, we note that the equity target terms $\sum_j x_{t.j}/M \cdot J_{(t-1)}(\mathbf{x}_{(t-1)})$ in $\tilde{Q}^{eqt}$ and $\sum_j x_{t.j}/\sum_j y_{t.j}(c_{t.j}^{(1)})$ in Q^{eqt} represent the statewide ratio of treatment centers per overdose death, which is less sensitive to each $x_{t.m}$ than the county-specific death estimates in the numerator. We, therefore, replace these ratios with year-specific constants R_t (estimated from observed data), making the optimization convex.

Second, even with this modification, the optimization is a mixed-integer quadratic program (MIQP), which is known to be NP-hard, and preliminary tests suggest intractable runtimes for even medium-sized states. To address this, we preestimate the term $\sum_j x_{t.j}/\sum_j c_{t.j}^{(1)}$ in the equality target (which represents the statewide number of centers per person, denoted P_t) from states' observed data, similar to R_t. This means that each county's equity (equality) targets are estimated as the product of their death counts (population) and the statewide center-per-death (center-per-person) ratios. These relaxations drastically simplify the problem and yield tractable runtimes. For the first MIQP run, we estimate R_{t_0} (P_{t_0}) as states' observed ratios of treatment centers per overdose death (per person) in year t_0 with subsequent R_t (P_t) estimates inflated by a factor of B_t/B_{t_0} (i.e., the proportional spending increase between year t_0 and year t). We then solve the MIQP a second time, using the implied R_t and P_t estimates from the first solution. We find that, by the second MIQP solution, the statewide ratios R_t and P_t nearly converge with values from the first and second MIQP runs differing by only 0.7% and 0.8%, respectively. Note that the optimization can be formulated without such relaxations using auxiliary variables (though it is still nonconvex), but runtimes quickly become intractable for even small states. Online Appendix EC.11 presents a validation analysis comparing the relaxed versus true solutions for a few states with tractable runtimes for which the average optimality gap is 0.22%.

3.5.6. Optimization Parameters and Software. We use the Gurobi 11 Python application programming interface to solve our optimization analyses for all 48 states

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in the continental United States, using an optimality gap of 0.1% (Kronqvist et al. 2019). The exp() function in $J()$ requires the use of auxiliary variables for $\hat{y}_{t.m}$ and $\log(\hat{y}_{t.m})$ connected with a piecewise linear exponential constraint (with a maximum relative error of 0.1%). We also use a piecewise linear approximation for the L2 equality norm with a maximum absolute error set to 0.30, which is 1% of the average county's squared deviation from its equality target. Finally, we require that all counties receive at least $L = 0.5$ times their observed 2016 allocations; we argue that this value helps to ensure that optimization solutions are politically palatable because any solution that removes the majority (> 50%) of centers in a large proportion of counties may be undesirable for policymakers and constituents (National Conference of State Legislatures 2020). The resulting optimization is convex and solves in a matter of minutes (at most) for each U.S. state and produces Pareto-optimal frontiers across the tested α values.

4. Results

Our final data set includes 37,019 county–year observations with nonmissing data and at least one recorded overdose death, spanning 2,581 counties, for an average of 14.3 years of data per county. These 2,581 counties represent 97.8% of the U.S. population (as of 2008, our sample midpoint).

4.1. Descriptive Statistics

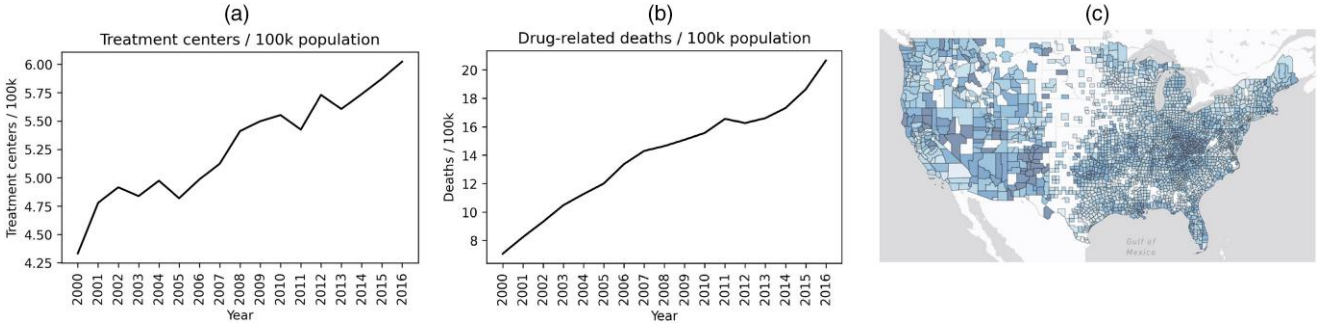
From 1999 to 2016, counties have an average of 6.8 treatment centers (SD = 19.0), 5.3 treatment centers per 100,000 residents (SD = 5.6), and 14.2 overdose deaths per 100,000 residents (SD = 10.6). Figure 1 plots average treatment center counts per 100,000 residents and overdose death rates from 1999 to 2016 along with counties' average overdose death rates from 1999 to 2016. A full descriptive statistics table for all variables appears in Online Appendix EC.12.

4.2. Econometric Model Results

Table 2 presents first stage regression coefficients from our econometric model. The F-statistic for Equation (1), which tests the strength of our instruments, is 97.21 ($p < 0.001$). At the average county population (96,356), the effect of parity-if-offered requirements for large-group plans on treatment center counts is −4.13, though this coefficient increases by 2.29 for each standard deviation (310,591) in population increase (note that these descriptives, with which population is standardized, come from our prefiltered data set of all U.S. counties). Thus, for counties with populations above 655,420, parity-if-offered mandates led to increases in SUD treatment centers. Thus, SUD parity-if-offered mandates may have led to a consolidation of treatment centers into higher population areas, similar to the consolidation observed in other healthcare sectors after Medicaid expansion (Cutler and Morton 2013). This trend is noticeable in the raw data; Figure A2 in Online Appendix EC.3 shows that states implementing parity-if-offered saw a subsequently higher concentration of treatment centers in large-population counties. Finally, the direction and significance of first stage effects remain if predicting treatment center density (i.e., centers/person or $x_{t.m}/c^{(1)}_{t.m}$) as the endogenous regressor, though only treatment center counts are predictive of overdose death rates in the second stage.

Table 3 lists coefficients from our second stage model, for which $R^2 = 0.4406$. Treatment centers significantly and beneficially impact county-level overdose death rates, $\theta = -0.00433$ ($p < 0.001$), with each additional center predicted to reduce death rates by $1 - \exp(-0.00433) = 0.43\%$. Note that if treatment center density ($x_{t.m}/c^{(1)}_{t.m}$) is included as an endogenous regressor, the coefficient for treatment center counts is almost unchanged ($\theta = -0.00434^{***}$), whereas treatment center density is insignificant. This both supports our use of treatment center counts as the focal predictor and suggests that treatment centers are more effective (in terms of number of deaths averted) in high-population counties as in Swensen (2015). In Online Appendix EC.5,

Figure 1. (Color online) Treatment Centers and Overdose Deaths Over Time and by U.S. County



Notes. (a) Treatment centers and (b) substance-related mortality per 100,000 residents from 1999 to 2016; (c) 2,581 U.S. counties included in the analysis, shaded based on their average overdose death rate from 1999 to 2016. Counts <10 are suppressed to comply with the CDC data use agreement.

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Table 2. First Stage Econometric Coefficients for Excluded Instruments

Coefficient	Variables	Value	*p*-value
$\dot{\beta}_1$	*Parity-if-offered*	−4.13**	0.008
$\dot{\beta}_2$	*Partial parity*	−0.044	0.991
$\dot{\beta}_3$	*Full parity*	0.055	0.998
$\dot{\beta}_4$	*Parity-if-offered × Population*	2.29***	0.000
$\dot{\beta}_5$	*Partial parity × Population*	−3.41	0.250
$\dot{\beta}_6$	*Full parity × Population*	−4.00	0.126

Note. Model includes all control variables, including county population. Population is standardized so that parity coefficients can be interpreted for the average-population county. $F = 97.21$.
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

we rerun our empirical model using Bartik-style shift-share instruments by instrumenting treatment center counts with counties' share of residential versus outpatient centers and the nationwide growth in each center type. The coefficient in this analysis is very similar ($\theta = -0.00461$) to our main analysis. Furthermore, Online Appendix EC.13 underscores the meaningfulness of this effect by showing that it persists beyond the one-year lag used for our main results. Finally, Online Appendix EC.14 presents a cost-effectiveness analysis based on these empirical results in which we compare the societal benefit of center openings with their cost. We find that each SUD treatment center generates more than $300,000 in annual societal benefits because of their impact on overdose deaths and other spillover effects after subtracting labor costs.

4.3. Optimal Treatment Center Allocations

We now present our optimization results, which reallocate counties' 2016–2019 treatment center budgets.

4.3.1. Overall Improvements in Death Rates and Treatment Inequity/Inequality. Figure 2 summarizes the predicted percentage changes in statewide SUD deaths (left), treatment inequity (middle), and treatment inequality (right) under our recommended treatment center allocations. The plot visualizes the performance of each α value as a percentage of the performance of states' actual allocations. Prioritizing SUD death reduction with intensity $\alpha \geq 0.5$ yields fewer SUD deaths than states' actual allocations, whereas solutions with $\alpha < 1$ yield lower inequity and inequality. Thus, solutions with $0.5 \leq \alpha \leq 0.9$ lead to reductions in all three metrics. Regarding the solutions' temporal dynamics, improvements in overdose death rates generally increase over time as more centers are added, whereas for $\alpha \leq 0.9$, improvements in inequality and inequity remain relatively stable and well below states' actual inequity/inequality values. As a robustness check, Online Appendix EC.9 presents summary results under alternate values of the fixed cost parameter $v^{(fixed)}$ (specifically, at 50% and 150% of its original value) and suggests that the improvements in Figure 2 hold even if the relative cost of center openings increases or decreases.

Table 3. Second Stage Coefficients

Variable	Coefficient	*p*-value
SUD treatment centers	−0.00433***	0.000
Population/100,000	−0.0525	0.409
Log(population)	−0.666	0.062
Per-capita income	0.00039	0.892
Employment-population ratio	−0.0124	0.138
Poverty rate	0.00279	0.691
Rurality	−0.00847**	0.008
Percentage of population 19 years or younger	−0.0286	0.216
Percentage of population 65 years or older	−0.00512	0.684
Log(opioid prescription rates)	0.0164	0.865
Medicaid coverage expanded	0.179***	0.000
Uninsured rate	0.0072*	0.024
Percentage paying w/ private insurance	0.395	0.176

Note. $R^2 = 0.4406$.
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

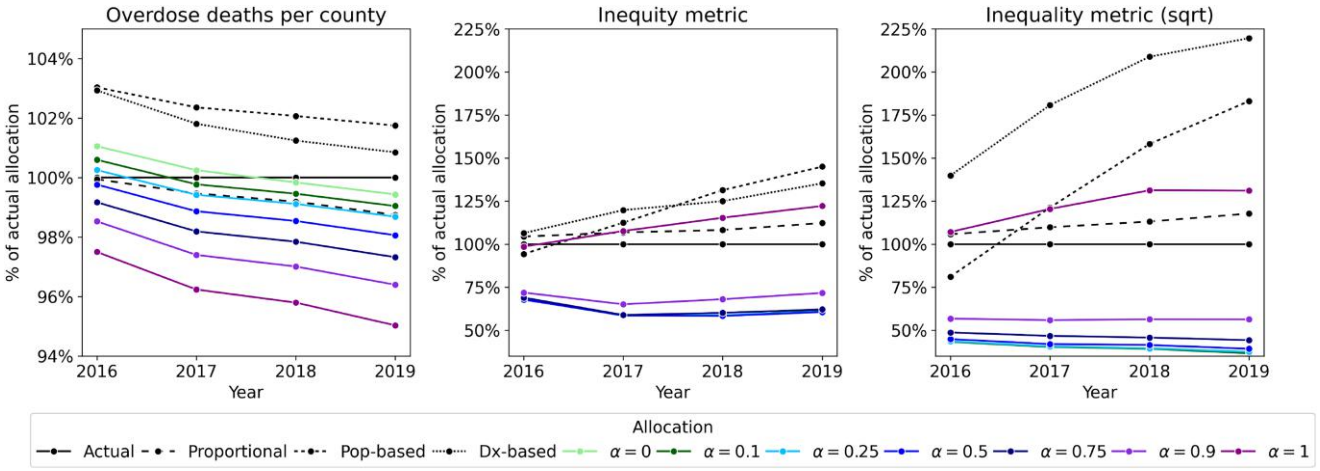
Regarding the proportional increase benchmark, simply increasing counties' treatment center counts by the same percentage reduces deaths but exacerbates existing inequities and inequalities. The population- and death-based benchmarks lead to increased death rates, inequity, and inequality. This is because centers in high-population counties are more expensive to operate than centers in low-population areas; thus, allocating funds according to population (or overdose deaths) leads to fewer centers per capita (or centers per death) in large counties than small counties. This exacerbates inequity and inequality and also reduces centers' overall effectiveness given that centers avert more overdose deaths in large-population counties.

Finally, the death rate improvements in Figure 2 remain steady even for decision year 2019 for which deaths are evaluated in 2020 (when overdose-related deaths increased considerably across the United States). Nevertheless, we also investigate whether a pessimistic approach, in which the optimization is solved using worst case predictions of yearly death rates, can improve solution quality. These results appear in Online Appendix EC.8; in short, we find that pessimistically forecasting yearly overdose rates μ_t to account for future rises in SUD mortality does not improve performance and produces slightly worse inequity/inequality scores (because assuming high death rates across all counties impacts their relative equity targets).

4.3.2. Comparison of Results by α and λ Values. We now present an in-depth comparison of how our results depend on both α and λ. Figure 3 plots improvements in death rates, inequity, and inequality for $\alpha \in \{0, 0.1, 0.25, 0.5, 0.75, 0.9\}$ and $\lambda \in \{0, 0.1, 0.25, 0.5, 0.75, 0.9, 1\}$ (we exclude $\alpha = 1$ because these results do not depend on λ). Each line in Figure 3 represents a single α value across multiple λ. This plot provides more insight

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Figure 2. (Color online) Predicted Percentage Improvements in Statewide SUD Deaths (Left), Treatment Inequity (Middle), and Treatment Inequality (Right) Compared with States' Actual Treatment Center Allocations Shown by Year

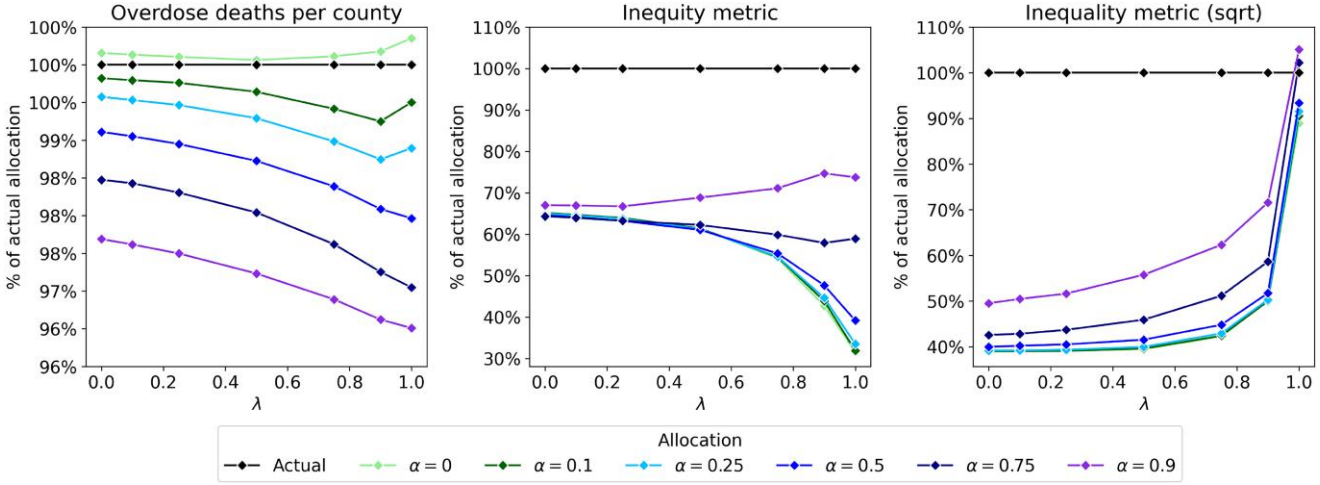


Notes. Allocations in years 2016–2019 are used to predict overdose deaths at a one-year lead, that is, for 2017–2020. Note that we report the square root of the inequality metric (i.e., the L2 norm rather than the squared L2 norm) for comparability with inequity (which uses an L1 norm).

into how setting $\alpha > 0$ contrasts with the special case of $\alpha = 0$.

Consistent with Figure 2, we find that setting $\alpha > 0$ results in fewer overdose deaths, greater inequity, and greater inequality than setting $\alpha = 0$. As α increases, each tested increment yields death rate improvements of roughly similar magnitude (Figure 3, left). Compared with $\alpha = 0$, increases in inequity and inequality when $0.1 \leq \alpha \leq 0.5$ are quite small on average, with more noticeable increases for $\alpha = 0.75$ and $\alpha = 0.9$ (Figure 3, middle and right). Increasing the equity weight λ leads to higher inequality (across α values) and generally leads to lower inequity when $\alpha < 0.9$. Interestingly, when $\alpha = 0.9$, the model is unable to meaningfully reduce inequity as λ increases (purple line in the middle plot of Figure 3). This is because overdose death rates are highest in low-population counties (note the near-significant effect of log-population in Table 3), meaning that inequity reduction requires placing more centers in small counties (which would increase death rates because centers are most effective in large counties). Thus, when α and λ are high, the model resolves this conflict by further reducing death rates at the expense of equity. In contrast, when λ is low, the model can achieve similar inequity levels regardless of α (see the left side of the middle plot of Figure 3). This is because of the inequity score's use of the L1 norm, which allows the model to trade off counties' deviations from their equity targets (i.e., moving one county away from its equity target can be offset by moving another county the same distance

Figure 3. (Color online) Improvements in Statewide SUD Deaths (Left), Treatment Inequity (Middle), and Treatment Inequality (Right) Under Various α and λ Values



Note. Note that λ is represented on the x-axis with improvements averaged across 2016–2019.

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toward its equity target). This is easiest when λ is low and counties are further from their equity targets, which allows the model flexibility in trading off counties' equity deviations. Thus, when λ is low, the same inequity score can be preserved even when α increases.

These results suggest that, when strongly prioritizing death rate reductions (e.g., $\alpha = 0.9$), decision makers may want to consider λ values of 0.5 or less (i.e., placing meaningful emphasis on equality) because the resulting increase in inequity relative to an $\alpha = 0$ case will actually be smaller than if λ were high (in which case, the death rate–inequity conflict would cause the model to further prioritize death rate reduction, yielding nonintuitive increases with λ). This further increases the attractiveness of the $\alpha = 0.5$ results, which yields lower death rates, inequity, and inequality than states' actual allocations, avoiding the nonintuitive conflict between death rate and inequity reductions.

4.3.3. State-Specific Improvements in Death Rates and Treatment Inequity/Inequality. To visualize state-specific results, we identify two optimization policies for each state:

- The minimal death rate policy, which corresponds to α_J, the highest alpha value that still lowers the λ-weighted inequity/equality average compared with states' actual allocations.
- The minimal inequity/inequality policy, which corresponds to α_Q, the lowest alpha value that still lowers SUD deaths compared with states' actual allocations.

Figure 4 plots reductions in statewide SUD deaths for each state's minimal death rate policy and reductions in inequity/inequality for their minimal inequity/inequality policies. Regarding the former, the figure suggests that SUD treatment center allocation could reduce predicted overdose deaths by as much as 14% (in the case of Illinois), while still reducing inequity and inequality in treatment access. As previously seen in Figure 2, potential improvements in inequity/inequality are quite large, reaching up to 80%. Notably, the states with the greatest death rate improvements are not necessarily those with the greatest inequity/inequality improvements; states' values from the top and bottom plots are correlated at $r = 0.07$, suggesting their ability to lower SUD death rates and inequity/inequality are largely unrelated to each other.

Of course, state-specific policies can be computed for any α value. Online Appendix EC.15 includes efficient frontiers for each state, which visualize their trade-offs between death rates and inequity/inequality. These trade-offs are summarized in Figure 5, which visualizes the absolute differences in each state's attainable death rate improvements (red) and inequity/inequality improvements (blue) between $\alpha = 0.1$ and $\alpha = 0.9$ (we exclude $\alpha = 0$ and $\alpha = 1$ here because we assume it is desirable to place at least some importance on both objectives). For states on the plot's left (with longer red/lighter-shaded lines), different α values yield large changes in statewide death rates (i.e., they have tall efficient frontiers in Online Appendix EC.15). For states on the plot's right (longer blue/darker-shaded lines), different α values yield large changes in inequity/inequality improvements (i.e., wide efficient frontiers). For states in the middle of the plot, their treatment center policies are relatively insensitive to α. Each state's label also includes $\hat{\alpha}$, the α value whose solution is closest to the state's actual treatment center allocation in 2019 (as measured by the Wasserstein distance between the actual and optimal proportion of centers assigned to each county) to provide insight into states' implied preferences.

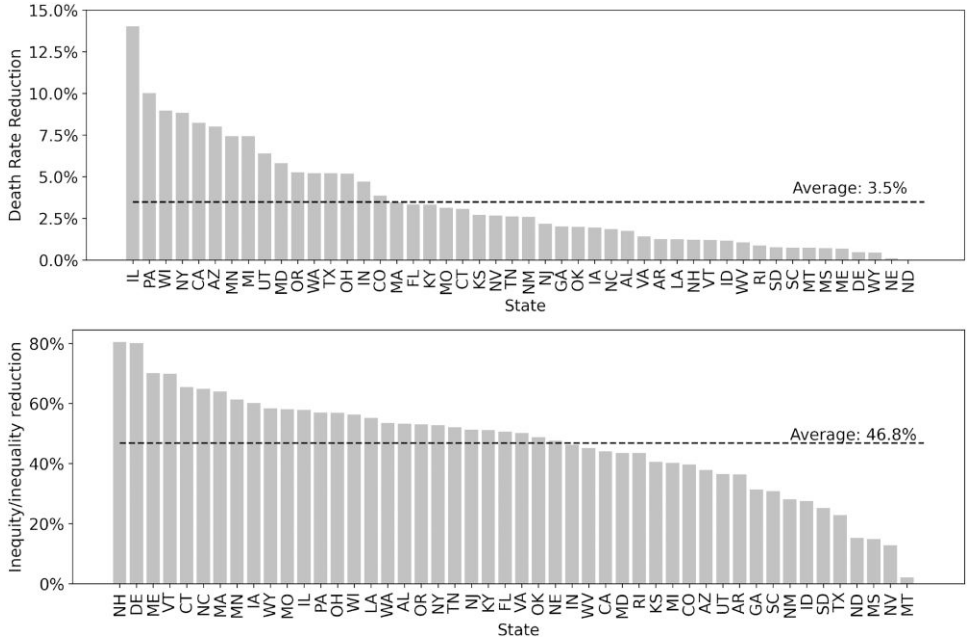
States where death rates are sensitive to α generally have at least one highly populous county; the correlation between states' death rate improvement spreads (length of the red/lighter-shaded lines in Figure 5) and their largest county's log-population is $r = 0.64$. Because centers are most effective in highly populous areas, moving centers into or out of such counties greatly impacts statewide death rates. Therefore, states with large urban areas, such as Illinois, Pennsylvania, New York, and California, must trade off large death rate improvements to reduce inequity/inequality. States where inequity/inequality are sensitive to α generally have high populations relative to their number of counties; the correlation between states' inequity/inequality improvement spreads (length of the blue/darker-shaded lines in Figure 5), and their population-per-county ratio is $r = 0.81$. Allocating centers to these states' most populous counties means directing resources away from several of the remaining counties, worsening overall inequity and inequality. In contrast, states with many counties (such as Texas and Georgia) can surge resources to high-population counties without affecting many smaller counties.

Finally, it is instructive to compare states' improvement spreads with their implied $\hat{\alpha}$ values, listed in the x-axis of Figure 5. For some states toward the left of the plot, observed treatment center allocations are most similar to the equity/equality-prioritizing $\alpha = 0$ policy. Yet these states' optimization results suggest that they can trade off relatively small increases in inequity/inequality for large gains in death rate reduction, and as such, it may be advantageous for these states to increase their emphasis on death rate reduction.

4.3.4. Temporal Dynamics of Treatment Center Allocation by County Type. We now characterize our solutions across all states. Figure 6 visualizes flows of center openings and closings from 2016 to 2019 under $\alpha = 0.1$ and $\alpha = 0.9$. The figures visualize center openings/closings for high versus low population counties and overserved versus underserved counties (yielding four county types). To define high- versus low-population counties,

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Figure 4. States' Predicted SUD Death Rate Reductions and Under Their Minimal Death Rate Policy (Top) and Predicted Inequity/Inequality Reductions ($\lambda = 0.5$) Under Their Minimal Inequity/Inequality Policy (Bottom), Averaged from 2017 to 2020, Compared with Actual Treatment Center Allocations

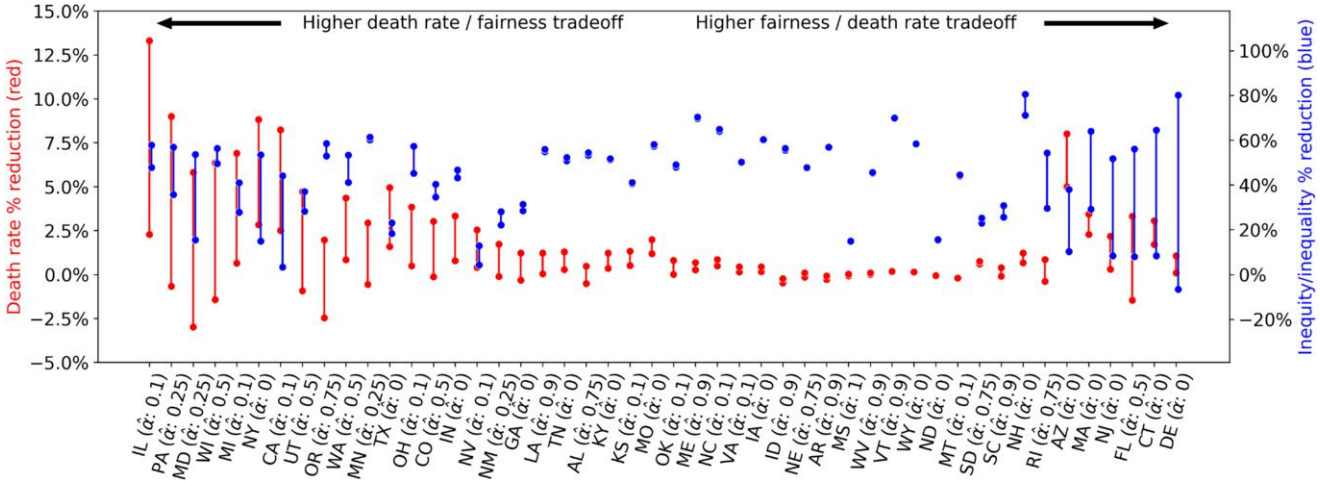


Note. Note that, for North Dakota, we set $\alpha_Q = 0.9$, which yields only a 0.03% increase in overdose deaths because inequity/inequality worsen considerably under $\alpha_Q = 1$.

we rank counties by population (ascending) within each state and identify the lowest ranked counties representing 50% of statewide population as low population and the others as high population. Overserved and underserved counties are defined by comparing each county's treatment center count to the value that would minimize its combined inequity/inequality score (derivation in Online Appendix EC.7). Each county type, therefore, represents roughly 25% of the statewide population.

According to Figure 6, the number of centers opened decreases over time as more funds are required to maintain/operate centers opened in previous years.

Figure 5. (Color online) Highest and Lowest Possible Death Rate Improvements (Red) and Inequity/Inequality Improvements (Blue) Across Each State's $\alpha = 0.1 - 0.9$ Efficient Frontier

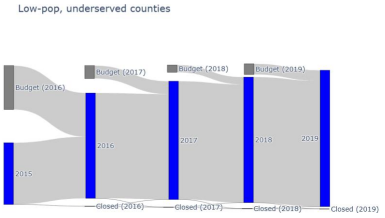


Notes. Note that inequity/inequality improvements are generally larger than death rate improvements, so values are scaled according to the maximum possible improvement for each. For states toward the middle of the plot, α has little effect on the improvements offered by either metric. State labels include $\hat{\alpha}$, the implied α value from the state's actual 2019 allocation.

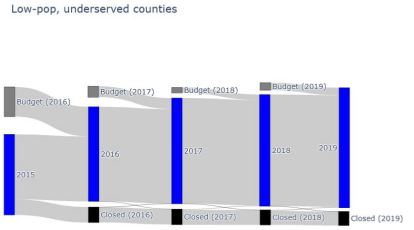
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Figure 6. (Color online) Sankey Diagrams Visualizing Overall Center Openings and Closings, 2016–2019

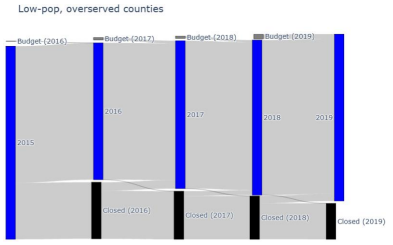
(a) Low-population, under-served, $\alpha = 0.1$. First node represents 2,687 centers in 2015.



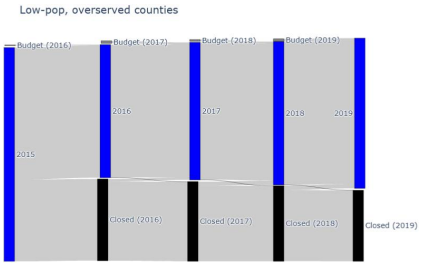
(b) Low-population, under-served, $\alpha = 0.9$. First node represents 2,687 centers in 2015.



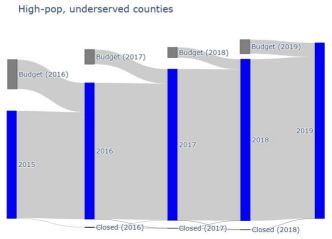
(c) Low-population, over-served, $\alpha = 0.1$. First node represents 5,607 centers in 2015.



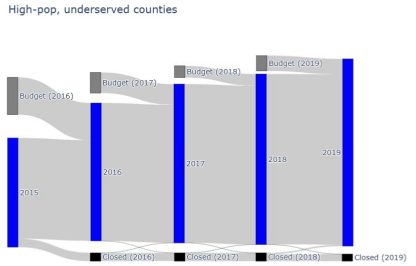
(d) Low-population, over-served, $\alpha = 0.9$. First node represents 5,607 centers in 2015.



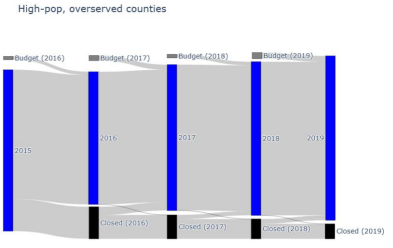
(e) High-population, under-served, $\alpha = 0.1$. First node represents 4,482 centers in 2015.



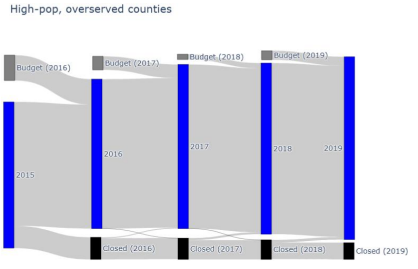
(f) High-population, under-served, $\alpha = 0.9$. First node represents 4,482 centers in 2015.



(g) High-population, over-served, $\alpha = 0.1$. First node represents 4,906 centers in 2015.



(h) High-population, over-served, $\alpha = 0.9$. First node represents 4,906 centers in 2015.



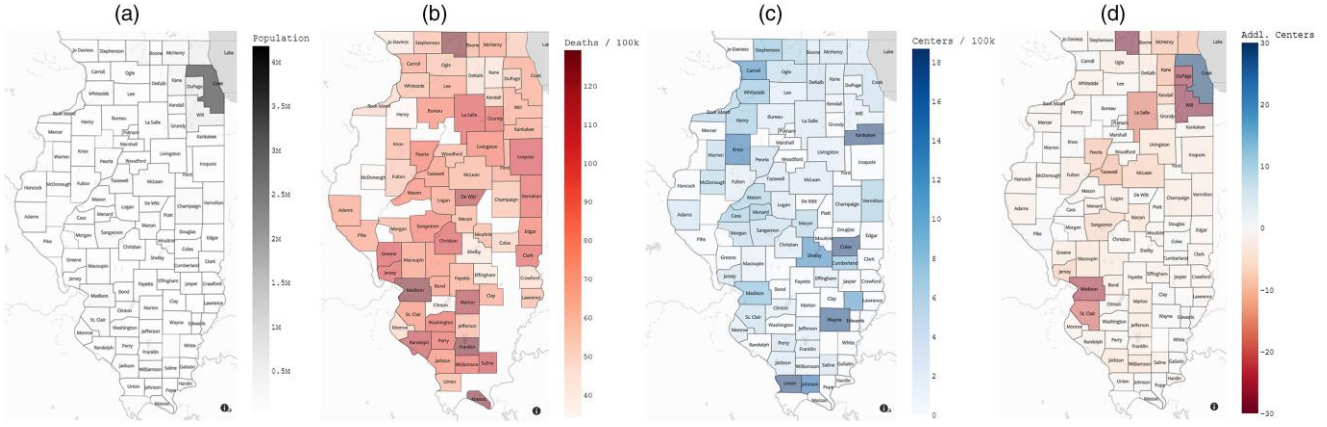
Notes. Blue nodes represent numbers of open centers during each decision year, gray nodes represent centers opened from yearly budgets, and black nodes represent closed centers. Flows from gray to blue nodes indicate centers opened, whereas flows from blue to black nodes indicate closed centers. Flows from black to blue nodes indicate centers being added to counties where they were previously closed and can be interpreted as center reopenings.

Several centers are closed (cease to be funded) in the first decision year. For both $\alpha = 0.1$ and $\alpha = 0.9$, several centers are closed in low-population, overserved counties because they contribute little toward death rate reduction or reducing inequity/inequality. High-population, overserved counties receive additional resources when $\alpha = 0.9$ (because such centers are predicted to reduce statewide death rate) but receive far fewer centers (and experience more closures) when prioritizing equity/equality. When $\alpha = 0.9$, more new centers are directed to high-population than low-population underserved counties given the centers' greater impact on statewide death rate in the former.

The reverse is true when $\alpha = 0.1$ because, when prioritizing equity/equality, increasing treatment center density is less expensive in low-population counties.

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Figure 7. (Color online) Optimization Solution ($\alpha = 0.9$) for Illinois



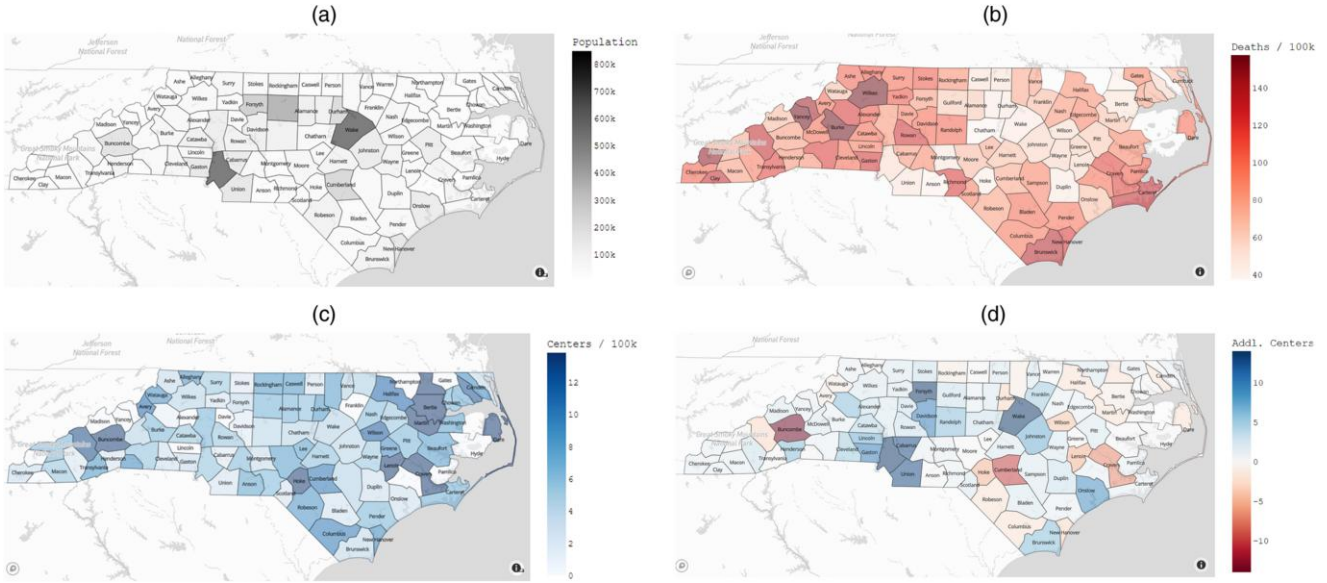
Notes. Panels (a)–(c) visualize county population, death rate, and centers per 100,000 residents (respectively); panel (d) shows optimal changes in treatment centers from 2015 (predecision year) to 2019. Death rates are shown from 2012 to 2016. Panel (d) is scaled to represent most counties' changes; the increase for Cook county is 151 centers. Death counts <10 are suppressed to comply with the CDC data use agreement.

Finally, note that, when $\alpha = 0.1$, some centers in overserved counties are reopened in later years once more funds become available. This is because previously overserved counties may find themselves underserved as the overall stock of centers increases though, importantly, it is advantageous to redirect funds from these centers in earlier years to level the playing field across all counties. Table A3 in Online Appendix EC.16 presents county characteristics for the five variables that most distinguish these four county types. We find that underserved counties have about 25% higher exposure to opioids than overserved counties, potentially compounding the consequences of insufficient treatment, and that the disparities in treatment center densities between underserved and overserved counties is greater for low-population counties than high-population counties.

4.4. State-Specific Case Studies

We analyze the recommended treatment allocation for Illinois, which sees the greatest potential for reduced death rates, under $\alpha = 0.9$ (which we assume is more desirable than completely ignoring inequity/inequality). Figure 7 plots county characteristics and the recommended change in centers per county between 2015 (predecision year) and 2019. As seen in panel (d), several centers are added to highly populous Cook county (which includes Chicago) and to rural counties in the

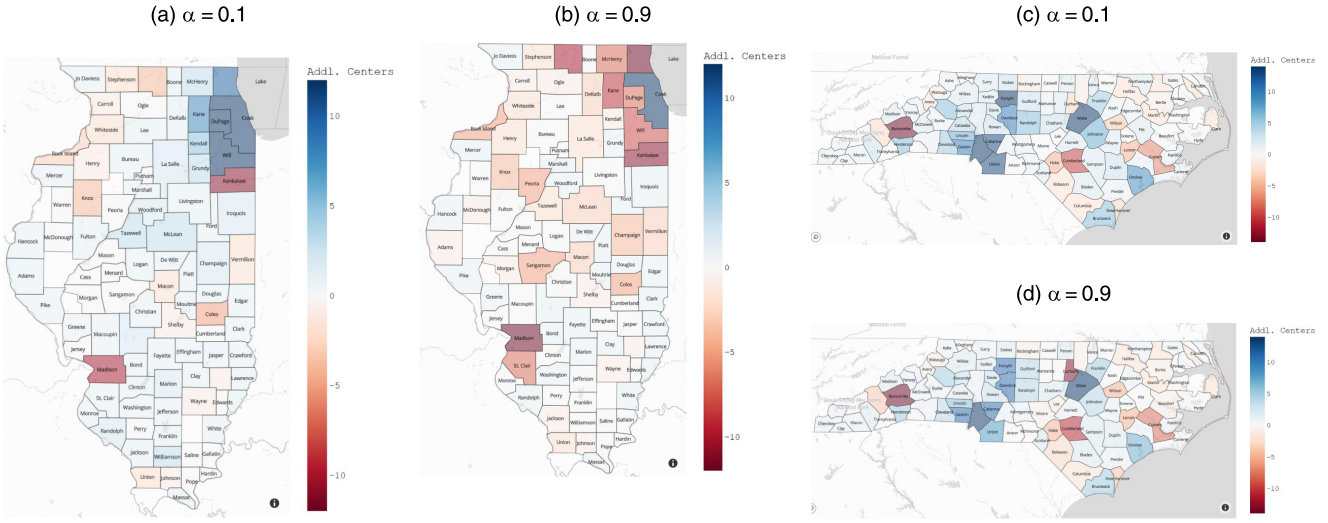
Figure 8. (Color online) Optimization Solution ($\alpha = 0.1$) for North Carolina



Notes. Panels (a)–(c) visualize county population, death rate, and treatment centers per 100,000 residents (respectively); panel (d) shows optimal changes in treatment centers from 2015 (predecision year) to 2019. Death counts <10 are suppressed to comply with the CDC data use agreement.

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Figure 9. (Color online) Additional Centers Allocated to Each County Under $\alpha = 0.1$ vs. $\alpha = 0.9$ for Illinois and North Carolina with $\lambda = 0.5$



south (which generally have few treatment centers and high death rates) as a relatively inexpensive way to improve inequity/inequality. To fund this, several centers are closed in the moderate-population counties surrounding Chicago and in the southwest (near St. Louis). These counties generally have either low death rates (Kane and Lake near Chicago) or high treatment center concentrations (Madison and St. Clair in the southwest).

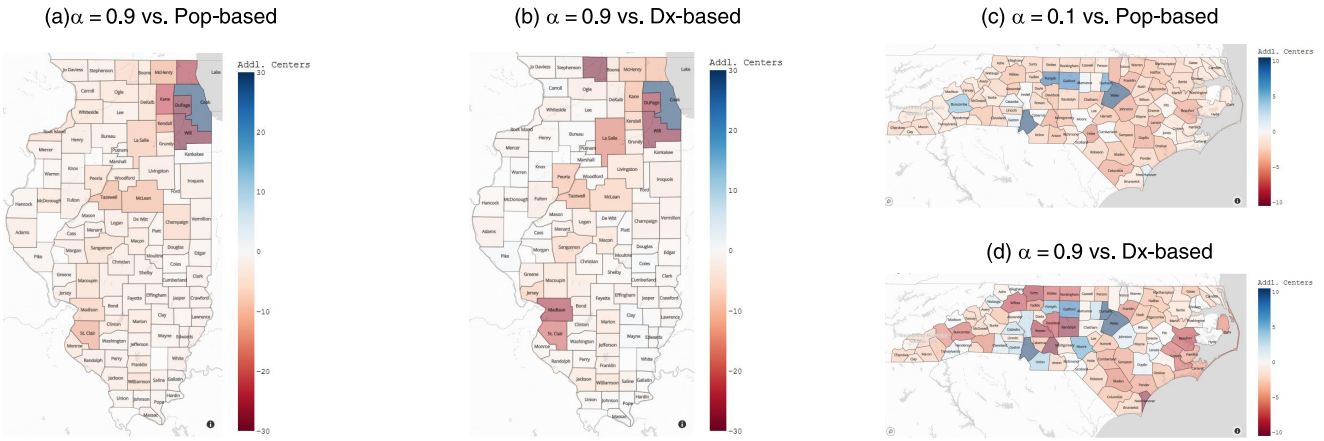
We now analyze North Carolina, which sees high potential for reduced inequity/inequality and serves as a rich case study because of its above-average number of counties (unlike higher ranked states in the bottom plot of Figure 4, which have very few counties). Figure 8 visualizes North Carolina's county characteristics and recommended changes in treatment centers for $\alpha = 0.1$. As seen in panel (c), many of North Carolina's rural eastern counties have relatively high center concentrations, whereas its more urban and suburban counties (Mecklenburg and Union in the Charlotte area, Wake in the Raleigh area, and to some extent Guilford in the Greensboro area) have relatively low treatment center concentrations. Under the $\alpha = 0.1$ allocation, these urban areas receive additional treatment centers, whereas centers are closed in several of the overserved eastern counties.

4.4.1. Adjusting Allocation Strategies with α and λ. Figure 9 shows optimal changes in treatment centers for Illinois and North Carolina under $\alpha = 0.1$ and $\alpha = 0.9$. For both states, a higher α directs centers to high-population areas. The increase to Cook county in Illinois is 151 centers (or 2.89 centers/100,000 residents) under $\alpha = 0.9$ but only 29 centers (or 0.55 centers/100,000 residents) under $\alpha = 0.1$. The increases to Wake (Raleigh) and Mecklenburg (Charlotte) counties in North Carolina are 51 and 41 centers (respectively) under $\alpha = 0.9$ but only 34 and 19 centers under $\alpha = 0.1$. Overall, α has a greater impact for Illinois than North Carolina as seen earlier in Figure 5, and its $\alpha = 0.1$ policy requires a steep trade-off in death rate reduction compared with $\alpha = 0.9$. In contrast, North Carolina's $\alpha = 0.1$ and $\alpha = 0.9$ policies are relatively similar and yield similar death rate and inequity/inequality improvements (Figure 5). This is likely because North Carolina's population is not dominated by a single urban area and because its largest cities are somewhat underserved and, therefore, receive additional centers under any α.

Online Appendix EC.10 presents Illinois' and North Carolina's results for $\lambda \in \{0.1, 0.9\}$. Illinois' differences under $\lambda = 0.1$ versus $\lambda = 0.9$ are small; Cook county receives a few more centers when $\lambda = 0.9$ because its share of statewide deaths (46%) exceeds its share of population (41%). North Carolina's largest counties (Mecklenburg and Wake) have a smaller share of statewide deaths than population (6.5% versus 9.7% for Mecklenburg and 5.5% versus 9.5% for Wake), therefore receiving fewer centers when equity is prioritized.

4.4.2. Comparison with Population- and Death-Based Benchmarks. Finally, Figure 10 visualizes how our case study states' solutions compare with the population-based and death-based benchmarks. Compared with these benchmarks, our solutions recommend higher center concentrations in urban areas, such as Cook County in Illinois and Mecklenburg and Wake Counties in North Carolina. This is because simply allocating funds according to population or overdose deaths leads to underinvestment in urban counties because of their higher center operating costs. This effect persists even if mainly prioritizing equity and equality as with

Figure 10. (Color online) Difference in Recommended Treatment Center Counts Between Optimal Solutions and Population-Based (Pop-Based) and Death-Based (Dx-Based) Benchmarks



Note. Colors are scaled to represent most counties' differences; the differences for Cook County in panels (a) and (b) are 209 and 190.

the $\alpha = 0.1$ solution for North Carolina. For both states, our recommended solution leads to lower death rates than the population-based and death-based benchmarks (Illinois: 97.7%, 114.4%, 111.8% of baseline, respectively; North Carolina: 99.5%, 100.6%, 100.8% of baseline, respectively). Similarly, both states see lower inequity under our recommended solution compared with the population-based and death-based benchmarks (Illinois: 48.4%, 149.5%, 123.7% of baseline, respectively; North Carolina: 48.3%, 85.3%, 88.5% of baseline, respectively) as well as inequality (Illinois: 36.0%, 158.1%, 149.4% of baseline, respectively; North Carolina: 22.1%, 98.2%, 129.6% of baseline, respectively).

5. Discussion and Conclusion

This is, to our knowledge, the first study to investigate optimal, sequential allocation strategies for SUD treatment resources. We formally incorporate allocation strategies' inequity and inequality alongside empirically based estimates of efficiency and do so in a sequential manner in which decision units' equity targets depend on prior decisions. We demonstrate how decision makers can tailor resource allocation to community preferences and characterize states' flexibility in balancing fairness and efficiency.

We find that states' existing levels of SUD funding can be reallocated in a manner that reduces overdose deaths, inequity, and inequality in treatment access compared with benchmark approaches. Importantly, simply increasing counties' SUD resources by a fixed percentage can exacerbate existing inequalities or inequities in treatment access. Similarly, simply allocating funding based on population or overdose deaths leads to underinvestment in high-population counties, which can also exacerbate inequity and inequality.

Our results suggests that SUD funding should be rebalanced as much as possible in the first decision year with centers closed in some counties and their funding redirected to others. Our results recommend closing several centers in low-population, overserved counties and centers in high-population, overserved counties if prioritizing equity/equality. New and redirected funding is generally spent on underserved counties with a greater share directed to high-population counties if prioritizing death rate reduction and low-population counties if prioritizing equity/equality. Thus, even though rural areas are disproportionately affected by SUD and by poorer access to healthcare, we find that directing resources to urban areas can be a rational part of SUD resource allocation. Another interesting finding is that centers may be reopened in previously overserved counties as counties' equity and equality targets increase over time.

Not all states benefit from our optimization in the same ways. States with large urban areas can reduce their death rates by surging centers to such areas. The degree to which this increases inequity/inequality depends on whether such areas are overserved or underserved; in New York, Manhattan County has more centers per capita (11.24 per 100,000 residents) than the statewide average (6.67) such that adding centers to Manhattan increases inequity/inequality (Figure 5). For Illinois, Cook County has 5.32 centers per 100,000 residents, close to the statewide average of 4.97, meaning that increases in Cook County do not detrimentally increase inequity/inequality. Our results for Illinois ($\alpha = 0.9$) recommend additional investment at the extremes of the population distribution: additional centers are recommended in Cook County (where centers can contribute to death rate reduction) and in smaller, underserved counties where increasing treatment

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access is relatively inexpensive. To fund these increases, centers are closed in moderate-population counties that generally have either low death rates or already sufficient treatment access.

Most states can substantially reduce inequity and inequality in their treatment allocations though they differ in their ability to trade off inequity/inequality with death rates. For states with high populations relative to their number of counties, equity/equality can be traded off for moderate death rate reductions (because centers can be removed from most counties to serve the higher population areas); for other states (those in the middle of Figure 5), meaningful death rate improvements are not as easily achieved, and therefore, all α values yield similar inequity-/inequality-minimizing policies. This lack of policy diversity can also occur for states such as North Carolina, where high-population urban areas are underserved (and, therefore, always receive additional centers).

Our models are missing data for some U.S. counties; future work should validate our findings with a complete county data set, perhaps within a single state. Future work might also model the heterogeneous effects of different treatment types. Our data set also lacks information on the utilization rates of treatment centers; if all treatment centers in a county operate at capacity, moving centers away from that county would have a more detrimental effect on than moving resources from a county with lower utilization.

Whereas we find that the number of SUD treatment centers has increased over time, future research might extend our model by quantifying the impacts (in terms of increased SUD death rates, inequity, and inequality) associated with varying levels of budget cuts being considered by policy makers. Such work might also consider the impact of facility closing costs and/or revenue. We focus only on the costs of center openings (not closings) because the literature differs as to how facility closings should be modeled; some papers consider salvage and/or rental income from closed facilities (Greenleaf and Harrison 1987), some consider closing costs (Wang et al. 2003), and some do not model closing costs/revenue because of such ambiguity (Verdonck et al. 2016). Our focus on opening costs also reflects our model's state-level perspective in which policymakers distribute funding (e.g., block grants) whose purpose is to increase access to SUD care. As such, it would be unusual for states to directly fund facility closures; similarly, sale or rental revenue from closed facilities would likely accrue to the provider rather than to the state. Nevertheless, future studies might explore how varying assumptions regarding center closure impact the results.

In line with other healthcare resource allocation work (Long et al. 2018, Jakubik and Feuerriegel 2022), our study uses a deterministic decision model. Yet future work could explore stochastic modeling of county overdose death rates using techniques such as dynamic programming or stochastic average approximation. For instance, future studies might model overdose deaths using Markov decision processes (MDPs) and solve for the optimal resource allocation policy across counties. Future studies might also consider the use of stochastic average approximation in which the resource allocation problem is solved across a finite number of sampled overdose death trajectories. Note that these modifications may require alterations to the inequity metric, which depends on county-level overdose rates. If overdose death rates are allowed to fluctuate stochastically, then counties' equity targets may need to be computed with respect to a moving average of prior overdose death rates to prevent large swings in counties' equity targets over time.

Altogether, optimization-driven allocation of SUD treatment resources can meaningfully reduce SUD-related deaths, reducing inequity and inequality in treatment access. Policymakers and healthcare providers can, therefore, leverage our data-driven allocation framework to identify underserved counties, improve treatment access, and reduce drug-related overdose deaths in their states.

Acknowledgments

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References

Ahmadi-Javid A, Seyedi P, Syam SS (2017) A survey of healthcare facility location. *Comput. Oper. Res.* 79:223–263.

Argyris N, Karsu Ö, Yavuz M (2022) Fair resource allocation: Using welfare-based dominance constraints. *Eur. J. Oper. Res.* 297(2):560–578.

Asamani JA, Alugsi SA, Ismaila H, Nabyonga-Orem J (2021) Balancing equity and efficiency in the allocation of health resources—Where is the middle ground? *Healthcare* 9(1257).

Ascension Recovery Services (2022) How much capital do you need to open an addiction center? Accessed March 2, 2025, https://www.ascensionrs.com/the-climb/the-cost-of-opening-an-addiction-treatment-center.

Baucum M, Khojandi A, Myers C, Kessler L (2023) Optimizing substance use treatment selection using reinforcement learning. *ACM Trans. Management Inform. Systems* 14(2):1–30.

Bertsimas D, Farias VF, Trichakis N (2012) On the efficiency-fairness trade-off. *Management Sci.* 58(12):2234–2250.

Bondurant SR, Lindo JM, Swensen ID (2018) Substance abuse treatment centers and local crime. *J. Urban Econom.* 104:124–133.

Buhat CAH, Lutero DS, Olave YH, Quindala KM, Recreo MGP, Talabis DAS, Torres MC, Tubay JM, Rabajante JF (2021) Using constrained optimization for the allocation of covid-19 vaccines in the Philippines. *Appl. Health Econom. Health Policy* 19(5):699–708.

Bureau of Labor Statistics (2024) Quarterly census of employment and wages. Accessed November 27, 2024, https://www.bls.gov/cew/downloadable-data-files.htm.

Cleveland Clinic (2024) Substance use disorder. Accessed March 2, 2025, https://my.clevelandclinic.org/health/diseases/16652-drug-addiction-substance-use-disorder-sud.

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Conover CJ, Bailey J (2020) Certificate of need laws: A systematic review and cost-effectiveness analysis. *BMC Health Services Res.* 20(1):1–29.

Cutler DM, Morton FS (2013) Hospitals, market share, and consolidation. *JAMA* 310(18):1964–1970.

Dave D, Mukerjee S (2011) Mental health parity legislation, cost-sharing and substance-abuse treatment admissions. *Health Econom.* 20(2):161–183.

Diderichsen F (2004) Resource allocation for health equity: Issues and methods. Accessed March 8, 2025, https://documents1.worldbank.org/curated/pt/451071468762912145/pdf/315940HNP0Chap1enRAforHlthEqtyFinal.pdf.

Drug Enforcement Agency (2024) Arcos retail drug summary reports. Accessed March 2, 2025, https://deadiversion.usdoj.gov/arcos/retail_drug_summary/arcos-drug-summary-reports.html.

Drug Rehab Agency (2022) Startup cost for an outpatient rehab center. Accessed March 2, 2025, https://www.drugrehab.agency/addiction-marketing-blog/startup-cost-outpatient-rehab.

Duan N (1983) Smearing estimate: A nonparametric retransformation method. *J. Amer. Statist. Assoc.* 78(383):605–610.

Edmond MB, Aletraris L, Roman PM (2015) Rural substance use treatment centers in the U.S.: An assessment of treatment quality by location. *Amer. J. Drug Alcohol Abuse* 41(5):449–457.

Eriksen MD, Rosenthal SS (2010) Crowd out effects of place-based subsidized rental housing: New evidence from the LIHTC program. *J. Public Econom.* 94(11–12):953–966.

Firouz M, Li L, Cobb B, Shao F (2023) On the equity-efficiency trade-off in food-bank network operations. *J. Oper. Res. Soc.* 74(12):2493–2514.

Frank RG, Beronio K, Glied SA (2014) Behavioral health parity and the affordable care act. *J. Soc. Work Disability Rehabilitation* 13(1–2):31–43.

Freedman M (2024) How to open a private medical practice. *Business News Daily* (January 16), https://www.businessnewsdaily.com/8910-opening-a-medical-practice.html.

French MT, Drummond M (2005) A research agenda for economic evaluation of substance abuse services. *J. Substance Abuse Treatment* 29(2):125–137.

French MT, Dunlap LJ, Zarkin GA, McGeary KA, McLellan AT (1997) A structured instrument for estimating the economic cost of drug abuse treatment: The drug abuse treatment cost analysis program (DATCAP). *J. Substance Abuse Treatment* 14(5):445–455.

Ghose R, Forati AM, Mantsch JR (2022) Impact of the covid-19 pandemic on opioid overdose deaths: A spatiotemporal analysis. *J. Urban Health* 99(2):316–327.

Greenleaf NE, Harrison TP (1987) A mathematical programming approach to elementary school facility decisions. *Socio-Econom. Planning Sci.* 21(6):395–401.

Gross CN, Brunner JO, Blobner M (2019) Hospital physicians can't get no long-term satisfaction—An indicator for fairness in preference fulfillment on duty schedules. *Health Care Management Sci.* 22:691–708.

He Q, Barkowski S (2020) The effect of health insurance on crime: Evidence from the Affordable Care Act Medicaid expansion. *Health Econom.* 29(3):261–277.

Hunter GP (2022) The intersection of harm reduction and zoning law: A growing divide between NIMBYs and public health initiatives. *Indiana Health Law Rev.* 19(2):481.

Jakubik J, Feuerriegel S (2022) Data-driven allocation of development aid toward sustainable development goals: Evidence from HIV/AIDS. *Production Oper. Management* 31(6):2739–2756.

Karsu Ö, Morton A (2015) Inequity averse optimization in operational research. *Eur. J. Oper. Res.* 245(2):343–359.

Kronqvist J, Bernal DE, Lundell A, Grossmann IE (2019) A review and comparison of solvers for convex MINLP. *Optim. Engrg.* 20:397–455.

Langabeer JR, Gourishankar A, Chambers KA, Giri S, Madu R, Champagne-Langabeer T (2019) Disparities between US opioid overdose deaths and treatment capacity: A geospatial and descriptive analysis. *J. Addiction Medicine* 13(6):476–482.

Lee MR, Sankar V, Hammer A, Kennedy WG, Barb JJ, McQueen PG, Leggio L (2019) Using machine learning to classify individuals with alcohol use disorder based on treatment seeking status. *EClinicalMedicine* 12:70–78.

Lien RW, Iravani SM, Smilowitz KR (2014) Sequential resource allocation for nonprofit operations. *Oper. Res.* 62(2):301–317.

Lishner DM, Levine P, Patrick D (1996) Access to primary health care among persons with disabilities in rural areas: A summary of the literature. *J. Rural Health* 12(1):45–53.

Long EF, Nohdurft E, Spinler S (2018) Spatial resource allocation for emerging epidemics: A comparison of greedy, myopic, and dynamic policies. *Manufacturing Service Oper. Management* 20(2):181–198.

Ma Y, Wang T, Zheng H (2023) On fairness and efficiency in nonprofit operations: Dynamic resource allocations. *Production Oper. Management* 32(6):1778–1792.

Marotta PL, Hunt T, Gilbert L, Wu E, Goddard-Eckrich D, El-Bassel N (2019) Assessing spatial relationships between prescription drugs, race, and overdose in New York state from 2013 to 2015. *J. Psychoactive Drugs* 51(4):360–370.

McIntosh J, Bloor M, Robertson M (2008) Drug treatment and the achievement of paid employment. *Addiction Res. Theory* 16(1):37–45.

McKay NL, Lemak CH, Lovett A (2008) Variations in hospital admin costs. *J. Healthcare Management* 53(3):153–166.

Medlock J, Galvani AP (2009) Optimizing influenza vaccine distribution. *Science* 325(5948):1705–1708.

Mishra M, Mathias T (2019) U.S. launches four-state study to find ways to reduce opioid overdose deaths. *Reuters* (April 19), https://www.reuters.com/article/business/healthcare-pharmaceuticals/us-launches-four-state-study-to-find-ways-to-reduce-opioid-overdose-deaths-idUSKCN1RU2EN/.

Nasir M, Summerfield NS, Oztekin A, Knight M, Ackerson LK, Carreiro S (2021) Machine learning–based outcome prediction and novel hypotheses generation for substance use disorder treatment. *J. Amer. Medical Informatics Assoc.* 28(6):1216–1224.

National Conference of State Legislatures (2015) Mental health benefits: State laws mandating or regulating. Accessed March 2, 2025, https://www.ncsl.org/health/mental-health-benefits.

National Conference of State Legislatures (2020) Improving rural health: State policy options for increasing access to care. Accessed March 2, 2025, https://www.ncsl.org/health/improving-rural-health.

Office of National Drug Control Policy (2022) National drug control strategy. Accessed March 2, 2025, https://bidenwhitehouse.archives.gov/ondcp/the-administrations-strategy/national-drug-control-strategy/.

Palmer W (2020) What is the cost of starting a medical practice? Wolters Kluwer. Accessed March 2, 2025, https://www.wolterskluwer.com/en/expert-insights/what-is-the-cost-of-starting-a-medical-practice.

Pardo B (2017) Do more robust prescription drug monitoring programs reduce prescription opioid overdose? *Addiction* 112(10):1773–1783.

ParityTrack (2021) State parity implementation survey. Accessed March 2, 2025, https://www.paritytrack.org/reports/.

Pullen E, Oser C (2014) Barriers to substance abuse treatment in rural and urban communities: Counselor perspectives. *Substance Use Misuse* 49(7):891–901.

Purrington K (2019) Tackling the opioid crisis: What state strategies are working? National Academy for State Health Policy. Accessed March 2, 2025, https://nashp.org/tackling-the-opioid-crisis-what-state-strategies-are-working/.

Rais A, Viana A (2011) Operations research in healthcare: A survey. *Internat. Trans. Oper. Res.* 18(1):1–31.

Rea D, Froehle C, Masterson S, Stettler B, Fermann G, Pancioli A (2021) Unequal but fair: Incorporating distributive justice in operational allocation models. *Production Oper. Management* 30(7):2304–2320.

Rosenblum D, Unick J, Ciccarone D (2017) An instrumental variables approach to estimating the effects of changes in the heroin market on overdose in the us. Technical report, Canadian Centre for Health Economics, Toronto.

Sommers BD, Baicker K, Epstein AM (2012) Mortality and access to care among adults after state Medicaid expansions. *New England J. Medicine* 367(11):1025–1034.

Stewart D, Gossop M, Marsden J (2002) Reductions in non-fatal overdose after drug misuse treatment: Results from national treatment outcome research study. *J. Substance Abuse Treatment* 22(1):1–9.

Substance Abuse and Mental Health Services Administration (2024) Substance abuse and mental health block grants. Accessed March 2, 2025, https://www.samhsa.gov/grants/block-grants.

Sun K, Sun M, Agrawal D, Dravenstott R, Rosinia F, Roy A (2023) Equitable anesthesiologist scheduling under demand uncertainty using multiobjective programming. *Production Oper. Management* 32(11):3699–3716.

Swensen ID (2015) Substance-abuse treatment and mortality. *J. Public Econom.* 122:13–30.

Tragler G, Caulkins JP, Feichtinger G (2001) Optimal dynamic allocation of treatment and enforcement in illicit drug control. *Oper. Res.* 49(3):352–362.

Verdonck L, Beullens P, Caris A, Ramaekers K, Janssens GK (2016) Analysis of collaborative savings and cost allocation techniques for the cooperative carrier facility location problem. *J. Oper. Res. Soc.* 67(6):853–871.

Wang Q, Batta R, Bhadury J, Rump CM (2003) Budget constrained location problem with opening and closing of facilities. *Comput. Oper. Res.* 30(13):2047–2069.

Warner BD, Leukefeld CG (2001) Rural-urban differences in substance use and treatment utilization among prisoners. *Amer. J. Drug Alcohol Abuse* 27(2):265–280.

Wen H, Cummings JR, Hockenberry JM, Gaydos LM, Druss BG (2013) State parity laws and access to treatment for substance use disorder in the United States: Implications for federal parity legislation. *JAMA Psychiatry* 70(12):1355–1362.

CORRECTION

In the online version of this article, "Reducing Overdose Deaths and Mitigating County Disparities: Optimal Allocation of Substance Use Treatment Centers" by Matt Baucum, Matt Harris, Lawrence Kessler, and Guanyi Luc (*Manufacturing & Service Operations Management*, vol. 27, no. 3, pp. 736–756), Matt Baucum's email address (page 736) was updated and his current affiliation was added to the Acknowledgments (page 754).

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